



A review of current research and prospects of fused deposition modelling: application, materials, performance, process variables, parameter optimization, and numerical study

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Abstract

Additive manufacturing is one of the most popular technologies for various engineering applications. Specifically, fused deposition modelling (FDM) is a primary additive manufacturing method that can deposit the material in a layer-by-layer manner. However, the performance of FDM-fabricated parts is affected by process parameters, material properties, and fabrication methods. This review study presents a comprehensive review of fused deposition modelling (FDM), concentrating on application, process parameters, parameter optimization, modelling, performance metrics, and numerical simulation. The main points of additive manufacturing and FDM are reviewed, highlighting their benefits, like design flexibility, cost-effectiveness, and suitability for environmental impact. The challenges of material limitation, process parameter optimization, and mechanical properties are addressed through literature findings. Noteworthy developments in modelling and optimization methods such as Taguchi, response surface methodology, artificial neural networks, fuzzy logic, genetic algorithms, and machine learning are discussed. Future research directions include enhancing material range, enhancing quality metrics, and advancing additive manufacturing methods. This review aims to provide valuable insight into current FDM research and its potential for innovation and sustainability.

Keywords Additive manufacturing · Fused deposition modelling · Parameter optimization · Numerical simulation · FDM materials · Mechanical properties

1 Introduction

Subtractive and formative procedures are the principles of part fabrication. Additive manufacturing, branded as 3D printing, is an advanced manufacturing method to manufacture components by adding material surface to surface, film by layer, line by line, and point by point to generate a

part. Based on the fundamental principle of forming a system, the material used, application, and energy sources, the additive manufacturing technique is categorized into fused deposition modelling, powder bed fusion, electron beam melting, and laminated object manufacturing. The main principle involved in the 3D printing method is the generation of a CAD model and control system, slicing the model into 2D cross-sections, and translating these 2D data to the AM machine to manufacture the physical product.

The fused deposition modelling (FDM) method, a commonly used AM method, is widely utilized to manufacture the part and has received the maximum consumer interest and innovation in the field of additive manufacturing. FDM is a non-stop thermoplastic filament that is liquefied and dropped by a printing head with a nozzle in a layer-by-layer manner, utilizing the G-code-specific printing parameter. In this method, a complex geometrical shape object with a net shape fabrication method that promotes the creation of new material with the least expensive manufacturing system, simple and fast fabrication approach, higher product cycle

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time, low cost, and used to print biodegradable materials such as polymers, polymer composites, and polymer blends for a sustainable environment that makes FDM valuable in various fields for the fabrication of single apparatuses in automotive, medical, aviation, education, and functional equipment.

Regardless of its benefits, FDM has problems that occur in the printing method. The fabrication process, quality, mechanical characteristics, and efficiency of the component cannot meet the requirements. Controlling thermal stress that leads to cracking and deformation during the deposition procedure is another major problem. Additionally, the internal surface of the component pores and microcracks is easily produced during printing. These problems arise due to conflicting process parameters such as slicing parameters, temperature parameters, and infill parameters, process parameter settings, material properties, production of products, and post-process treatments.

Considering these problems, numerous studies focus on experimental and numerical studies for enhancing manufacturing methods such as hybrid manufacturing techniques, utilizing the blending of pure material and addition of fiber material, improving forming quality and mechanical properties. Other studies utilize statistical methods, such as the design of experiments, Taguchi approach, and fractional factorial designs to determine the influence of the different process variables on the quality objective. Some studies are concerned with advanced mathematical modelling and optimization techniques, including response surface methodology, artificial neural networks (ANN), fuzzy logic, grey relation, genetic algorithms (GA), and machine learning methodologies to optimize the FDM process parameters to enhancing part characteristics, component quality, and verifying the experimental result.

To reverse the limited materials availability, accuracy, and quality, anisotropic mechanical characteristics, limited application in large component production, mass production, printing time, clogging, and void formation also need extensive research.

This study seeks to explore the current advancements and future potential of fused deposition modelling (FDM), a widely used additive manufacturing technique. It emphasizes key areas such as its diverse applications across industries, the materials employed, and the performance of the method under various conditions. The investigation also delves into process variables, examining how they influence the quality and efficiency of the fabricated parts. Additionally, the study focuses on parameter optimization through experimental and numerical approaches, striving to enhance the characteristics of printed components. This study is organized into seven sections: Section 1 provides an overview of various studies on 3D printing, materials used in FDM, performance

metrics, FDM process variables, modelling and optimization methods, and numerical simulation. In addition to these, the study shows the relevance and effectiveness of the methods utilized in the current trends and identifies research gaps in additive manufacturing. In the second section, the study identifies the additive manufacturing process. The third section describes the performance metrics, process parameters and printed specimens applied in FDM. In the fourth section, the design of the experiment, parameter modelling, optimization technique, and numerical studies used in FDM are discussed. In Section 5, significant contributions of the work and challenges of the FDM process are discussed. Section 6 focuses on insights into future research directions, and the last section draws some conclusions from the paper. Achieving these study goals, the paper contributes to the development of advanced manufacturing methods, the development of smart materials, the advancement of FDM technology, and the enhancement of environmental impact.

2 Additive manufacturing process

2.1 History of FDM/3D printing

Charles Hull invented 3D printing in 1986, and its beginnings lay in the invention of the inkjet printer in 1976 [1]. Since its invention, FDM was invented by Scott Crump in 1989 and improved throughout the subsequent years, while the printers were commercially legalized in 1990, and Wake Forest Institute of Regenerative Medicine used the technology to print 3D scaffolding for organ augmentation. It processes like hot glue and starts as a desktop size, while no matter the size is designed. The machine only has gears, stepper motors, nozzle(s), and belts. Its growth and popularity of FDM are based on various research works that have been done to increase productivity without sacrificing quality and durability. The FDM machine is developed based on a Cartesian coordinate robot, and the motion of the printer is in the x, y, and z directions. Numerous FDM machines are designed, such as conventional printing, flexible FDM system, CP printer, in-situ fusion mechanism, 3D compaction printing, dual extruder, ideal five-axis CLFDM with curved layers, five-axis 3D printer, and multi-axis 3D printer [2].

Supports affect the surface quality during the process in a three-axis machine. Thus, now mobile machines and multi-axis machines have been developed to find the printing path and building dimension, whereas this machine eliminates obstacles such as staircase defects in the conventional FDM process [3]. The first FDM has printed the thermoplastic materials, including PLA and ABS. Over a year, the FDM printed material increased to include all thermoplastic materials and composites. Numerous composites, such as ceramic

matrix composites, polymer matrix composites, and metal alloy composites, are printed in FDM [4]. Polymer composites, known as fiber-reinforced polymer (FRP), consist of fibers and resins. FRP and its derivatives are printed in FDM with high stiffness and good mechanical performance. For enhancing strength, reducing weight, reducing shrinkage, and minimizing warpage of the printed part for different applications, nowadays, novel materials such as polyvinyl alcohol, geopolymers, and Sn-15Pb and Pb40-Sn alloys are also printed in FDM [5, 6].

In addition to increasing the mechanical properties and thermal characteristics of the FDM part, it is necessary to reduce lead time, reduce costs, lower energy consumption, and decrease material waste. Thus, researchers find numerous parameters in this technology have affect the printed part quality. Therefore, it is necessary to obtain the optimum elements, for example, raster thickness, raster gap, nozzle diameter, nozzle temperature, build orientation, layer thickness, infill pattern, and bed temperature [7, 8].

2.2 Types of additive manufacturing processes

According to the American Society for Testing and Materials (ASTM), 3D printing methods are grouped into seven main categories based on the material-forming processes. These are vat photopolymerization, material extrusion, material jetting, binder jetting, direct energy deposition, sheet lamination, and powder bed fusion [9–11]. The utilization of each method is nominated by considering availability, desired feature size, type and state of material, and its intended application.

3D printing methods are classified as raw material melting and non-melting methods. Extrusion manufacturing, extrusion deposition, multiphase jet solidification, and fused deposition modelling are types of material melting techniques, while robocasting, pressure-assisted micro-syringe, solvent-based extrusion free-forming, direct-write assembly, and low-temperature deposition are grouped under the method without material melting. On the contrary, laminated object manufacturing (LOM), solid forming and fused deposition modelling (FDM) are solid-state processes. Digital light processing (DLP), stereolithography apparatus (SLA), and material jetting (MJ) are liquid-based processes, and selective laser sintering (SLS), electron beam melting (EBM), and selective laser melting (SLM) are powder-based processes [12, 13].

Direct energy deposition, powered bed fusion, and selective melting are popular for metal 3D printing, which utilizes powder, metal, and wire as a raw material and uses laser arcs, electron beams, and other high-energy beams as a heat source to print a part. Mostly, for metals, laser melt deposition with powder feeding is commonly used [14, 15]. A

liquid photopolymer is a vat through targeted light-activated polymerization to produce printed parts. Thermoplastic materials such as PLA and ABS are mainly processed in FDM, while slurry materials such as liquid photosensitive polymer are used in photo solidification stereo forming, and sheet materials, including plastic and film, are used in layered solid manufacturing. The different techniques are classified according to the discrete-accumulation principle, material type, and energy source; detailed information is shown in Fig. 1.

The path planning research on FDM is better than that of direct energy deposition (DED), which is costly. However, DED can manufacture lightweight parts from metals [16–18]. In addition to FDM, SLA, extrusion-based bio-printing, inkjet, and poly-jet are the widely used additive manufacturing techniques [19, 20]. Studies [21] found that the SLA method offers clear advantages, making it a hopeful option for producing pure ABS and ABS-blended materials with enhanced mechanical characteristics. The same study [22] found a similar elastic modulus, with quantitative values around 2.2 GPa for both FDM, DLP, and SLA on ABS thermoplastic material. Regarding the UTS, the SLA has proven higher results, and DLP found higher overall elongation and higher total energy absorption. Currently, manufacturing sectors are approaching the utilization of a hybrid of conventional manufacturing processes with additive manufacturing, known as combined additive manufacturing, to minimize cost, time, and enhance product quality [23–25].

2.3 Fused deposition modelling (FDM)

Among the versatile additive manufacturing (3D printing) techniques, fused deposition modelling (FDM) is also known as fused filament fabrication (FFF) [29, 30]. FDM is an extrusion-based method to fabricate objects through hot thermoplastic filaments that are melted and extruded through a heated nozzle. During printing, the distance between the nozzle and the bed should be at a standard distance and constant throughout the bed. A large variety of filament materials, usually thermoplastics and composites, are considered. During the design of fused deposition modelling (FDM), material possibilities, extruder technologies, and temperature resistance were considered. The different printing methods are shown in Fig. 2.

In the process of printing 3D objects, the CAD model is designed using CAD software and saved in a 3D printer-comprehensible (STL (stereolithography) and VRML (virtual reality modelling language)) file format. Finally, the file is sliced, and the G-code and other printing parameters are arranged to print the materials. The steps involved in 3D printing of an object are shown in Figs. 3 and 4 [34, 35].

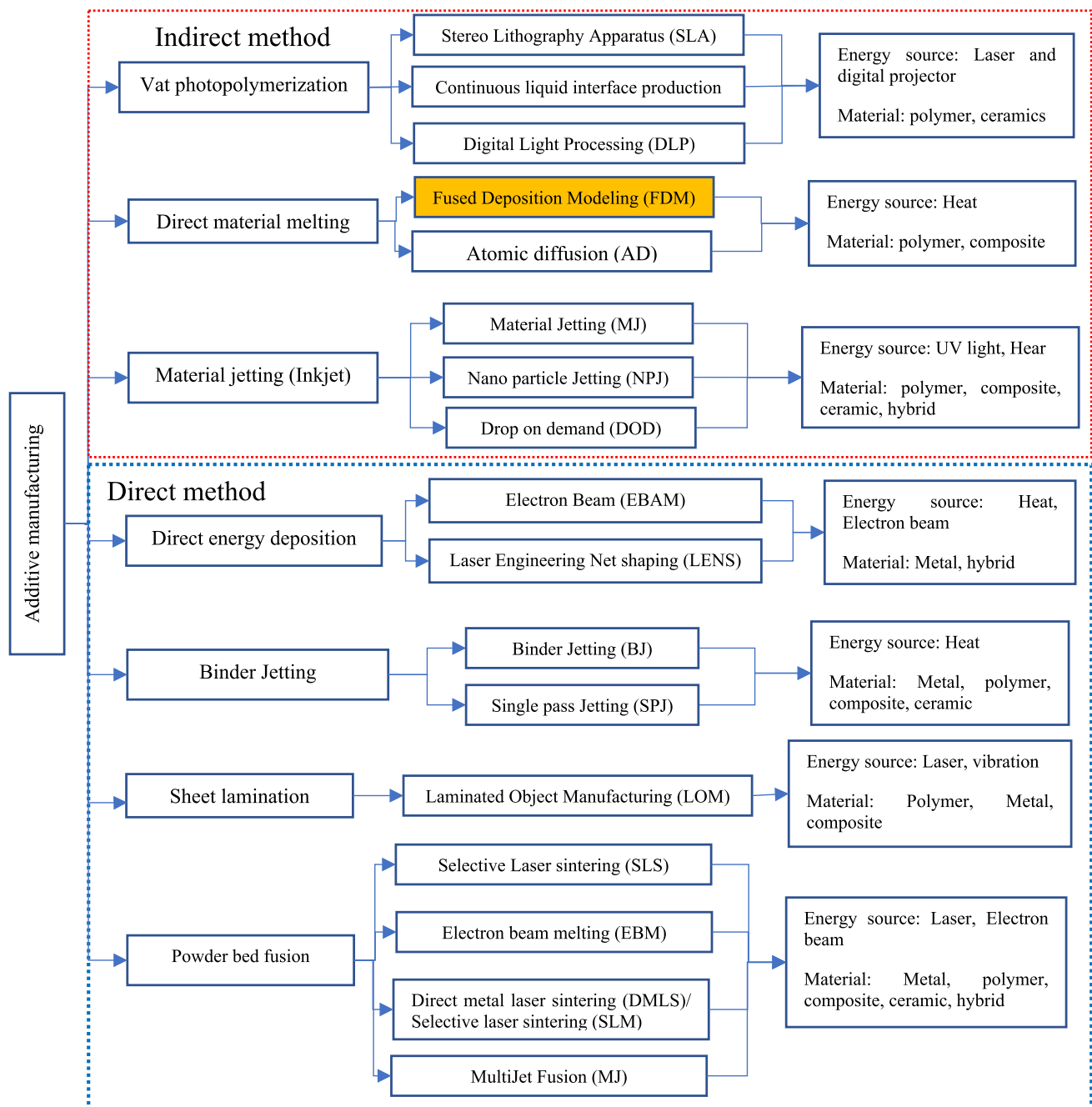


Fig. 1 Classification of additive manufacturing [26–28]

The 3D printer of the FDM method has a process parameter that consists of infill density, infill patterns, layer thickness, raster angle, build orientation, and temperature of extrusion, nozzle and bed. In addition to these, the machine has its parameters, including bed calibration and nozzle diameter [40, 41]. The parameters and potential metrics used in FDM are included in Fig. 5.

2.4 Applications of FDM

Additive manufacturing technology, especially FDM, has a wide range of applications and development prospects in various fields. It is used in various sectors, including industrial manufacturers, biomedical applications, aerospace industries, automobiles, electronics, construction

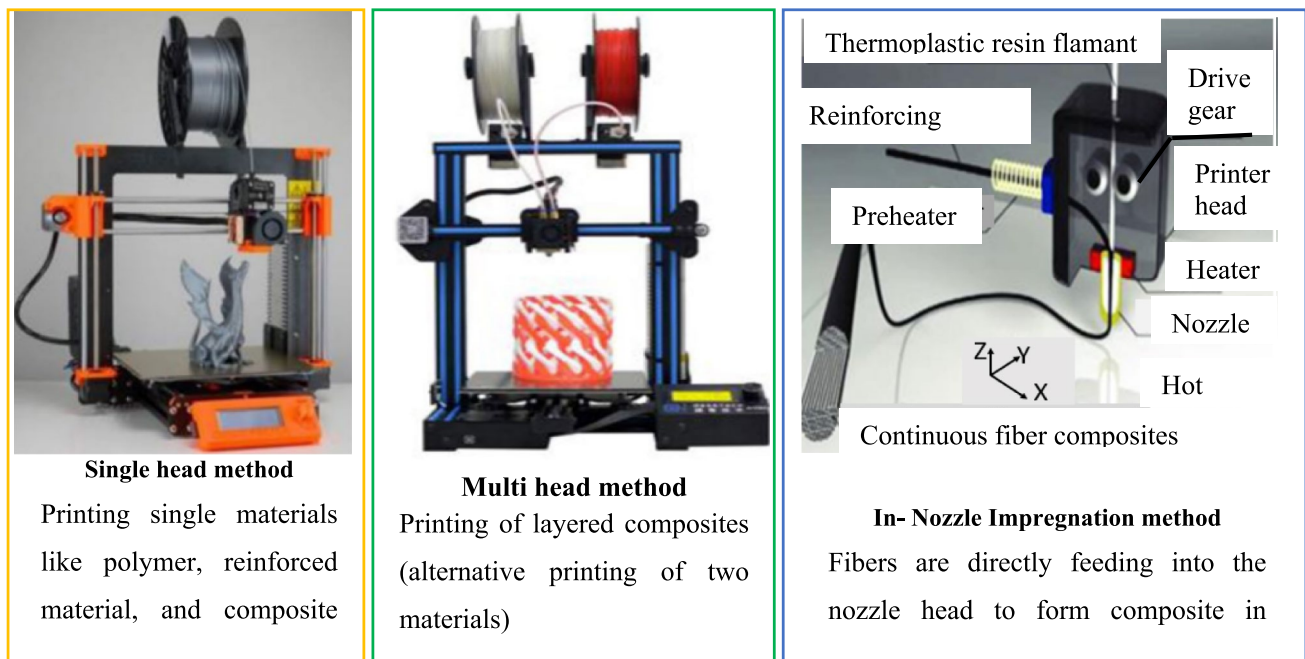


Fig. 2 Single, multi, and in-nozzle impregnation FDM method [31–33]

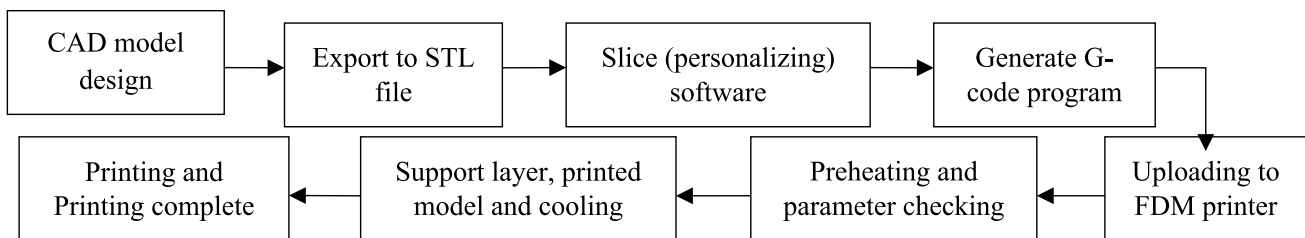


Fig. 3 The process flow of FDM

sectors, and researchers [45–47]. It is a rapid prototyping method for different functional components, lightweight products, multi-material products, complex geometry, medical components, and high-end equipment due to its low cost compared to conventional production methods [48]. The method is an incremental manufacturing technique that is flexible, reduces tool investment costs, optimizes materials usage, and reduces cycle time compared to conventional manufacturing methods [49].

FDM is used in the fabrication of implant material for medical records [50], tissue engineering, and the manufacturing of scaffolds, surgical guides, and artificial bones [51, 52]. Automobile components such as interior trim, dashboard parts, grilles, wheel covers, and exterior body panels were made by FDM [53–55]. The technology is not only limited to these applications but is also utilized in engineering

and other fields for the fabrication of casting models, personalized lamps, circuit components, printing pods, and mechanical components [56–58].

2.5 Materials printed in FDM

The most common materials that are printed using FDM 3D printers are selected based on their mechanical properties, thermal properties, and their specific applications (Fig. 6). Thermoplastics are lightweight, inexpensive, sustainable, and recyclable materials, which make FDM versatile, while metals, thermoset polymers, and ceramics are less commonly used in FDM due to their complexity in the production process [59]. Biodegradable thermoplastic materials are employed for the design, fabrication, simulation, and mechanical characterization of 3D-printed

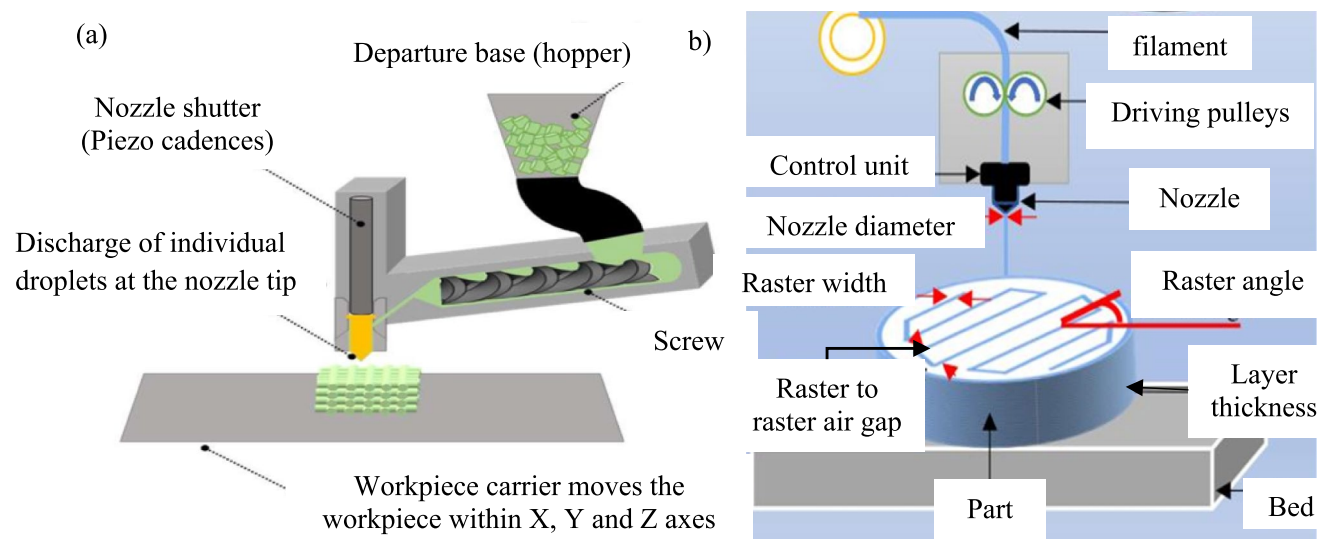


Fig. 4 Schematic principle of the 3D printing techniques: **a** Arburg plastic free forming droplet deposition modelling (APF DDM), and **b** fused deposition modelling (FDM) [36–39]

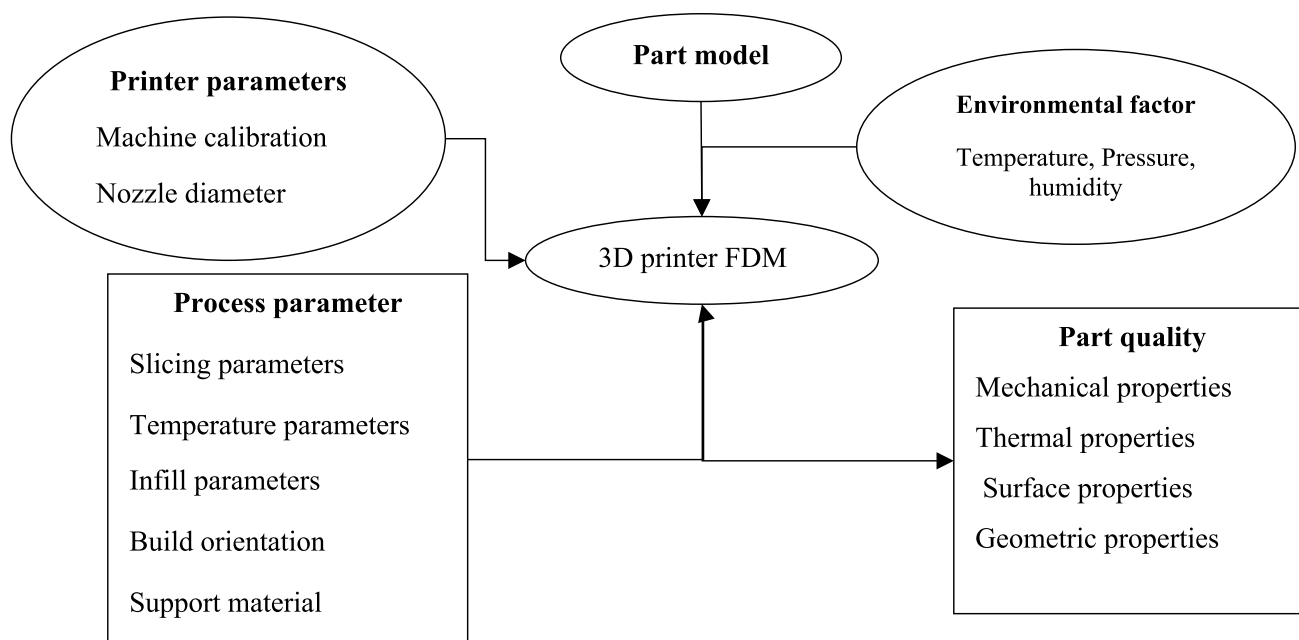


Fig. 5 Overview of the process parameters of FDM [26, 42–44]

Fig. 6 Common polymers used in the FDM process [64, 65]

Application	Semi-crystalline	Amorphous
High performance	PE, PP	PLA, PMMA, PVC
Engineering	PA, PBT, POM, PET	ABS, PC, PET-G
Commodity	PEEK, PPS, PVDF	PEI, PPSU, PSU

Performance decreases

components in specific fields. Thermoplastic materials, including acrylonitrile butadiene styrene (ABS), polycarbonate (PC), polyethylene terephthalate glycol (PETG), nylon, polycarbonates (PC), TPU, and polylactic acid (PLA), are among the most widely utilized materials in FDM 3D printing [60, 61].

Novel blended materials made from pure thermoplastic, reinforced materials, and their composites can replace commercially available materials and have resulted in increased strength and durability, as well as improved flexibility and resistance at high temperatures for fabricating high-quality parts for specific applications [62, 63].

2.5.1 Acrylonitrile butadiene styrene

Acrylonitrile butadiene styrene (ABS) is one of the most widely used thermoplastic polymer materials in FDM [66]. It is an amorphous copolymer composed of three primary monomers: acrylonitrile (15 to 35%), butadiene (5 to 30%), and styrene (40 to 60%) [64, 67]. Acrylonitrile enhances thermal and chemical resilience, along with surface hardness; butadiene enhances toughness and impact resistance, whereas styrene improves processability, rigidity, and overall strength of the material. The three copolymers have numerous properties, including lightweight, ease of printing, excellent toughness, heat resistance, impact resistance, rigidity, and relatively low cost. ABS has relatively good strength, acceptable thermal shrinkage, low glass transition temperature, dimensional stability, and the ability to perform functional products [66, 68]. It can withstand sunlight, rain, temperature, and environmental factors without significant degradation. It is useful for numerous applications across various industries, making prototypes of products for testing and validation before mass production, manufacturing automotive components (interior trim, dashboard parts, grilles, wheel covers, and exterior body panels), electronic devices (computer consoles, phone cases, consumer electronics), functional components (domestic materials, kitchenware, household appliances, storage containers, and decorative items), piping, education, and research (research laboratories, experimentation purposes, materials science). However, ABS tends to warp during printing, resulting in poor dimensional accuracy and lower layer adhesion. A medium nozzle temperature of 240 °C improves layer adhesion and reduces stringing, a bed temperature of 100 °C improves warping, deformation, and softening. A printing speed ranging from 45 to 50 mm/s enhances stringing and layer bonding, while a layer height of 0.2 improves surface roughness. Additionally, acetone vapor smoothing, and the application of adhesives such as glue stick, ABS slurry and using an enclosure further enhances layer bonding.

2.5.2 Polylactic acid

Polylactic acid (PLA) is among the biodegradable plastics, which is environmentally eco-friendly, widely available and decomposes easily without polluting the environment [69]. PLA is a thermoplastic monomer derived from renewable resources such as corn starch, sugarcane, tapioca roots, petroleum products, and other starchy plants [70]. It has interesting properties such as being nontoxic, bioactive, reduced ductility, and a renewable material, which is gaining popularity due to its sustainability [71]. It is widely used in fused deposition modelling (FDM) due to its easy printability and biodegradability. Compared to other thermoplastics, PLA offers low weight, higher tensile strength, lower warping, and lower odor. The melting point ranges between 170 and 190 °C, which requires a lower temperature and less energy to print, eliminating the need to heat the bed during printing and a negligible shrinkage effect after cooling [72]. A minimum layer thickness (0.05–0.1 mm), build plate temperature of 60 °C, extrusion (nozzle) temperature of 210 °C, printing speed of 60 mm/s, with higher infill density, produces a good printed part. The adhesion of the PLA printed part is enhanced by increasing the extruder temperature to 220 °C, and the warping is improved by increasing the bed temperature from 50 to 60 °C. Layer bonding and surface roughness are enhanced by minimizing the printing speed to 40 mm/s [73]. It is recommended to use 100% cooling fan and infill density greater than 50% with grid and honeycomb pattern to increase mechanical strength [74], moisture absorption to enhance brittleness, heat sensitivity, UV light exposure, and mechanical stresses and the UV resistance improved by UV-resistant coating, print in dark colors, and storing properly away from light exposure. PLA is commonly used for prototyping, educational projects, food packaging, compostable bags, containers, phone cases, and in biomedical applications for medical implants, where high quality and precision are essential [75].

2.5.3 Polyethylene terephthalate glycol

Another widely used material for the 3D FDM method is polyethylene terephthalate glycol (PETG). It is a modified variant of polyethylene terephthalate that is a better polymer for 3D printing. The addition of glycol to PET lowers the melting temperature to make it more user-friendly. PETG is durable, easy to print, UV-resistant, but it has poor adhesion, which creates strings between the layers and lowers fatigue resistance [76]. It has higher tensile strength, high ductility, relative flexibility, higher thermal stability, and excellent resistance to chemicals and temperatures, and it

is less brittle than PLA [77]. During printing PETG material, a printing temperature of 235 °C and a bed temperature of 80 °C minimize part deformation, reduce degradation, and increase adhesion. Furthermore, a printing speed of 50 mm/s and a layer height of 0.2 mm enhance strength, surface quality, and lower warpage. Enclosure is needed for consistency, while a dryer can avoid the moisture [78]. The lower anisotropy of the PETG is enhanced by optimizing build orientation, infill density, nozzle temperature, layer thickness, and print speed. It has been used in food containers, bottles, mechanical parts, and biomaterials for a variety of medical applications such as tissue engineering, orthopedics, neurology, gynecology, and surgery [79].

2.5.4 Nylon

Nylon is known as linear polyamide (PA), and is a family of petroleum-based synthetic polymers. It is generally used for FDM 3D printer filament fabrication. PA6 grade is more commonly used than the other grades [80]. Nylon has excellent wear and impact resistance, thermal resistance, and mechanical strength, while it can easily be exposed to moisture, which leads to poor quality and warping, and bed adhesion [81]. A smaller cooling fan, lower layer height (0.1 mm), adding an enclosure, and lower printing speed enhance bonding, strength, and reduce warping. In addition, bed temperature of 70 to 100 °C and higher printing temperature (190 to 350 °C) minimize warpage, while utilizing PET sheet and adhesives such as Garolite, PEI sheet can improve adhesion [82]. Before printing, nylon filament is heated with a temperature range of 80 to 90 °C for 5 h to remove moisture. A higher flow rate also produces a good printed part. Nylon is more flexible and durable compared to those of ABS. Some applications of nylon include textiles, fishing nets, packaging, precise and complex geometric shapes, consumer electronics, dental implants, joint implants, socks and fishing nets, and industrial gears [83].

2.5.5 Polyethene terephthalate (PET)

Polyethene terephthalate (PET) is a polyester family of synthetic polymers and is a popular material for FDM [84, 85]. PET is commercially available in a variety of forms, such as PETP, PETG, GPET, and PETT. Secondary microplastic production of PET from waste is recycled through mechanical methods and chemical recycling, including ammonolysis, alcoholics, and glycolysis, which produce plastic debris that affects the quality of the part [86, 87]. PET has excellent physical and chemical properties, including higher strength, durability, chemical resistance, and recyclability, which is good for the FDM process [88, 89]. The optimal extrusion (nozzle) temperature of 245 °C, glue stick, and a heated bed temperature of 80 °C utilized to prevent warping and

adhesion. The printing speed close to 30 mm/s and optimum layer tackiness of 0.25 mm can improve printing quality, surface quality, and adhesion. The bed adhesion can be used like a PET sheet glue stick to prevent warping and first-layer failure. To avoid the bubbling and poor surface quality, the filament has been dried at a temperature of 65 °C for 5 h. It is a widely used polymer in the production of water bottles, food packaging, trays, and flexible packaging [90]. In addition, PET is used for packaging in beverage industries, which accounts for 44.7% of single-serve beverage packaging [91]. This reuse of recyclable polymers is beneficial in the production of concrete products and wood-polymer boards. The production of PET reduces the environmental footprint by taking steps towards a sustainable future.

2.5.6 Thermoplastic polyurethane

Thermoplastic polyurethane (TPU) is a flexible filament, resistant to abrasion, grease, and oil. Thermoplastic polyurethane boasts a shore hardness of 95 and provides a clean, easy printing experience [92]. TPU is a strong yet bendable filament with excellent layer-to-layer bonding that prevents any layer separation. This flexible filament is semi-transparent and has a rubber-like appearance. Compared to other elastic filaments, TPU is easy to print, low-cost, has superior strength, abrasion resistance, toughness, shock absorption properties, durability, flexibility, chemical resistance, impact resistance, and elasticity [93]. The printing parameters, including lower bed temperature, slower printing speed, lower retraction speed, and fine layer tackiness, can improve the printing quality of TPU. The layer delamination of the TPU can be enhanced by reducing cooling and increasing the nozzle temperature. The material can absorb moisture, so it is needed to store it in a dry box and use drying before printing to enhance print quality and an ambient temperature of 20–30 °C is recommended during printing. A larger nozzle temperature with a larger nozzle diameter of around 0.4 mm and a higher flow rate can improve flow and adhesion, while a direct printing extruder prevents buckling and smooth printing [94, 95]. The material is widely used in various applications, including personalized customization, clothing design, aerospace, medical devices and surgical tools, non-pneumatic tires, shock absorbers, gaskets, and elastomeric parts [96].

2.5.7 Polyether ether ketone

Polyether ether ketone (PEEK) is an organic thermoplastic polymer of the polyaryletherketone family. It has excellent mechanical and chemical characteristics, a high strength-to-weight ratio, and higher strength than PLA and ABS. PEEK is used in human prostheses due to its potential bone-healing property. However, due to its high melting temperature and

high viscosity, it is exposed to deformation and crystallinity effects [97]. The melting point of PEEK is 343 °C. The effect of residual stresses and considerable deformations of the printed PEEK part can be minimized by adjusting the extrusion temperature to 467 °C, infill angle of 21°, and printing speed of 20 mm/s [98]. In addition, a small printing layer thickness of 0.1 mm with the same properties in the above parameter reduces internal defects, strengthens infill filaments, improves the density of PEEK, and reduces surface roughness [99]. Furthermore, a printing speed of 5 mm/s, a layer thickness of 0.1 mm, a printing temperature of 395 °C, and an extrusion strand width of 0.44 mm enhance the strength and crystallinity of PEEK. Annealing also improves the strength of the PEEK, crystallinity, and relaxation of residual stress [100].

2.5.8 Polycarbonate

Polycarbonate (PC) is an innovative thermoplastic material that has excellent mechanical properties, making it the strongest filament available in FDM 3D printers. It has a glass transition temperature of 150 °C, which is printed at temperatures of 310 °C and above, while it absorbs moisture, making defects in the printed part [101]. It is commonly used in designing future insulators, fire resistance, stability of its physical properties at high temperatures, and food containers [102]. The PC polymer has higher tensile strength in breakage, while elongation decreases compared to polypropylene and polyethylene polymers, as polycarbonate polymer acted as a brittle material for resistance to shock, whereas hardness, whose values were close to the three types of polymers [103]. Process parameters, including nozzle temperature of 270 °C, adding enclosure, layer height of 0.2 mm, and annealing, enhance the mechanical strength and improve the warpage of the printed PC.

2.5.9 Blended materials

PC-ABS material exhibits 24% higher elastic modulus than ABS and 16% higher compared to PC. In addition, PC-ABS exhibits 30% higher natural frequencies in CF end conditions and 26% higher in CC end conditions, while a nozzle temperature of 250 °C for ABS and PC-ABS, while nozzle temperature of 270 °C for PC with a bed temperature of 80 °C for ABS and PC-ABS, and 110 °C for PC were utilized [104]. The other parameters are constant for all printing specimens.

Study [105] found that the shape memory performance and mechanical properties were enhanced when ABS was blended with PETG. Common printing parameters, including printing speed of 250 mm/min, nozzle temperature of 235 °C, layer thickness of 0.45 mm, and nozzle diameter of 0.6 mm, were utilized. A tensile strength of 32.48 MPa

was found with a weight of PETG-ABS (30–70), due to the higher strength of ABS than PETG. The better shape memory effect of PETG also enhances the shape memory effect of PETG-ABS (70–30) than PET-ABS (30–50).

Higher tensile strength found at concentric patterns when polycarbonate (PC) was blended with ABS at a PC/ABS ratio of 70/30 than 60/40 using a layer height of 0.4–0.6 mm, printing temperature of 270–290 °C, printing speed of 30–50 mm/s, nozzle diameter of 0.4–0.8 mm, infill density of 100%, and infill pattern of concentric, linear, and wiggle [106]. A lower layer height (0.4 mm) and higher printing speed (50 mm/s) provided better quality, while the infill pattern had a higher contribution to the response, and the blend proportion could affect the deformation.

The mechanical test results revealed that adding TPU to ABS decreased the strength, storage modulus, and the glass transition temperature of ABS, while increasing the formability and elongation of the samples [107]. The crack initiation decreased from 568.4 N for ABS, while for ABS80, ABS60, and TPU, they are 335.3, 123.2, and 84.5 N, respectively, which shows the behavior of ABS changes from brittle to ductile. In addition, TPU reduces the size and number of voids and holes between the printed layers. During printing, a nozzle diameter of 0.8 mm and infill density of 100%, head thickness of 200 µm, and bed temperature of 65 °C were used. The printing temperature for ABS and ABS-TPU was 240 °C, and TPU was 230 °C.

2.5.10 Reinforced materials

Adding reinforcing fillers is one of the methods used to enhance the mechanical, electrical, chemical, and thermal properties of pure materials, reduces the price of the end products and has positive affects mechanical, electrical, and thermal properties [108] depends on the material type, the filler choice, and target functionality. The fillers are available in particle-reinforced, laminated, and fiber-reinforced forms, which are introduced to increase the composite strength [109].

Reinforcing elements are classified as synthetic fiber, short fiber, and continuous fiber. Specifically, carbon materials (carbon black, graphene, nanotubes, carbon fibers) are grouped in synthetic fibers. Continuum reinforced composites are popular in various applications due to their excellent mechanical properties [110]. The addition of fillers such as carbon materials, metals, ceramics, glass, and other fibrous fillers such as hemp, kenaf, jute, cellulose, flux, bamboo, cotton, and wood used to reinforce the structure and improve mechanical properties specifically load-bearing capacity (strength, stiffness, thermal conductivity, and warpage) compared to other reinforcements [111].

When reinforcing materials are incorporated into filaments for 3D printing, particle sizes can range from nanometers to micrometers, while filaments are commonly 1.75 mm or 2.85 mm in diameter. For example, wood particles in a composite filament may have sizes around 237 μm , while some nanoparticles used in reinforcement can be in the 20–60 nm range. Common 3D printing filaments include PLA, ABS, TPU, and nylon, and these can be modified with reinforcement materials. Natural fibers with a size less than 100 μm prevent blockings. However, flaxseed fiber with a 1 mm size is used as a fiber that can be successfully printed [112, 113]. In addition to the type, size, and distribution of the reinforcement material, the processing method and base material affects the quality of printed parts. Glass fiber has good visual appearance and improved mechanical properties and stability, while metal, wood, and stone filled filaments are found to have a good surface finish than carbon and glass fiber. Conductive and magnetic filaments offer unique properties that exhibit a specific surface finish [114].

Glass fiber at the top layer with continuous PLA filament sheet core, and carbon fiber bottom layer with epoxy resin, has superior mechanical performance compared to another configurations. A resin with thickness of 1.5 mm and fiber thickness of 1.5 mm was utilized. During printing a bed temperature of 80 to 100 $^{\circ}\text{C}$ and nozzle temperature of 100 to 180 $^{\circ}\text{C}$ with a nozzle diameter of 0.4 mm were used. Glass fiber (top) with FDM carbon filament sheet and carbon fiber (bottom) with addition of resin was found to have a maximum compressive strength of 53.06 MPa [115]. The addition of banana spathe biochar into ABS improved mechanical properties up to 26%. The highest fatigue life counts were observed for 2 wt% of biochar. Thus, the addition of banana spathe biochar in to ABS improves load-bearing effect, wear stability, and fatigue resistance, while having a minor effect on water absorption properties, which are more used in structural, automotive, defense, and medical applications [116].

From a 40-nm diameter of spherical groundnut shell-derived biocarbon of 50% in weight with PLA, it is founded a promising thermomechanical property exhibiting 39.8% increases in the tensile strength and 17.5% enhancement in tensile modulus than that of 25% and 75% [117].

The incorporation of fiber materials such as wood, cotton, Fe_2O_3 , zeolite, PCL composite, and carbon-based fibers in PLA significantly enhances mechanical properties and thermal resistance, which increases the applications and sustainability features of printed parts. In addition, blending materials including nanoscale clays and nanofillers enhances strength and stability and is cost-effective [118].

Carbon X (Gen3) with PLA6 carbon-reinforced nylon filament was printed with a nozzle diameter of 0.4 mm, an extrusion width of 0.4 mm, and an infill density of 20%. The study found that the incorporation of carbon filament at a

printing speed of 65 to 75 mm/s, with extruder temperature of 240 to 250 $^{\circ}\text{C}$, enhances the module's strength to 26.08 MPa and elongation 7.87% [119].

PETG reinforced with 20% carbon-fiber (CF) was printed with a layer thickness of 0.2 mm, infill density of 100%, nozzle diameter of 0.4 mm, extrusion temperature of 250 $^{\circ}\text{C}$, and bed temperature of 70 $^{\circ}\text{C}$ for PETG and CF-PETG. The printing parameters achieve a robust interlayer bonding. Due to the increase in material stiffness when adding CF to PETG, the Young's modulus is enhanced compared to pure PETG. The short carbon fiber-PETG does not affect the mechanical properties, while lowering elongation at break and surface roughness in CF-PETG, which reduces the ductility [120].

The incorporation of short carbon fiber with PLA fabricated in FDM enhanced the tensile strength when printing a layer thickness between 1.15 and 1.5 mm, while the flexural strength is improved when printing a layer thickness between 0.8 and 1.35 mm with a constant infill density of 100%. And also the addition of short carbon fibers in nylon induces microstructure change and promotes interlayer de-bonding due to voids and pores as per the study [121].

An increase in short carbon fiber content from 15 to 35 wt% decreases the nylon melt flow of (SCF-PA). A maximum tensile strength and flexural strength of 101.43 MPa and 173.16 MPa, respectively, are found at 25 wt% compared to pure nylon. A 35 wt% enhances the impact strength to 6.02 kJ/m^2 , which increases 38.1% compared to pure nylon. The study considers the same carbon size of 7 mm diameter with a printing temperature of 300 $^{\circ}\text{C}$, printing speed of 100 mm/min, infill density of 100%, bed temperature of 80 $^{\circ}\text{C}$, and layer height of 0.2 mm [122].

A wood dust fiber was treated using alkali, silane, and NaOH-silane. The mesh size ranges from 300 to 200 μm . The filament was extruded with a nozzle temperature of 160 to 200 $^{\circ}\text{C}$, and a 1.75 mm filament was fabricated [123]. The wood-PLA is employed in building constructions, mainly in structural applications, while carbon-PLA offers higher strength, light weight, and durability, which is used in the automotive industry. The wood-PLA offers higher mechanical properties than carbon-PLA [124].

A nozzle diameter of 0.6 mm, an extrusion temperature of 230 $^{\circ}\text{C}$, a bed temperature of 45 $^{\circ}\text{C}$, the layer thickness of 0.5 mm, and a printing speed of 35 mm/s found a better quality for the three conditions, including without magnet, magnet below, and magnets around the print area of PLA. The tensile strength rises from 13 to 14.04 MPa when 15 wt% of iron magnets are applied during printing. The sample printed with a magnet increases adhesion strength than without a magnetic field, while it weakens the adhesion strength of the printed part [125].

For a glass fiber reinforced-PLA (GFRP) composite axial compression strength of 16130.50 N and a maximum specific

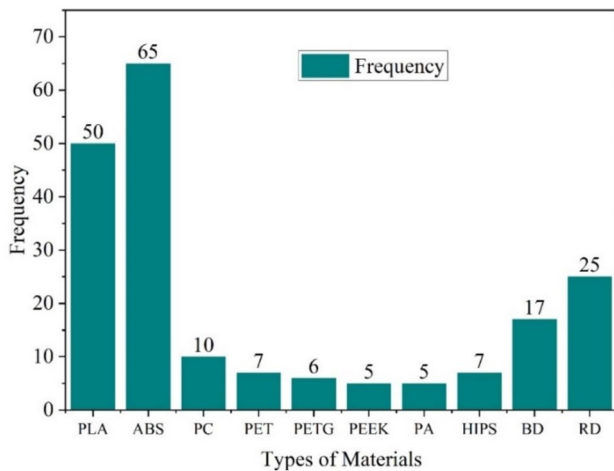


Fig. 7 Different types of polymer material for the FDM method as per the study (BD is blended and RD is reinforced)

energy absorption were founded at 262.18 J with PLA-GFRP 0.42%. An epoxy resin bisphenol with fiberglass of ER 2400 TEX is used. The density increases to 1.2 for PLA, to 1.7 for GFR and to 1.9 for GFRP. The addition of 0.41 wt% to 0.51 fiberglass material to PLA increases mechanical performance of printed part. A maximum bending force of 939.33 N is founded for PLA-GFRP 0.42%, while a minimum bending force of 218 N was obtained for PLA-GFRP 0.42% with a thickness of 4.89 mm [126]. The frequency of the material used in FDM is illustrated in Fig. 7.

3 Performance metrics and process parameters in FDM printed specimen

3.1 Performance metrics

Mechanical characteristics of components made using FDM additive manufacturing techniques are tested considering ASTM standards. Based on the imposed loads on the parts, the specimen with designated standards for flexural, impact, and tensile strength testing of parts is manufactured.

3.1.1 Review on the mechanical properties of pure biodegradable material

During mechanical testing, the maximum failure load, elongation at break, percent elongation, and tension were the areas of interest for different scholars.

The hexagonal lattice structure had a bending strength of 45 MPa and shear strength of 8 MPa, 13% and 24% higher than the plain lattice structure, respectively, with an extrusion temperature of 200 °C, a bed temperature of 60 °C, printing speed of 50 mm/se, and layer height of 0.2 mm

during printing of PLA. The triangular lattice structure also had a tensile strength of 28 MPa, which is 6% higher than the plain lattice structure. The plain lattice structure demonstrated a 17% increase in impact strength, which is 278 J/m higher than the circular structure [127]. A PLA material was printed at a bed temperature of 60 °C, extrusion temperature of 200 °C, printing speed of 35 mm/min, 22 layers with constant longitudinal and vertical orientation, printing time of 4.5 h, and layer thickness of 0.2 mm. After printing, the test specimens were heat-treated at 55 °C, 65 °C, and 80 °C for 5 h. The result discovered that the heat treatment of PLA material to 65 °C has increased its tensile strength by 35% without causing any deformation, improving the structural integrity, while decreasing porosity [128].

3.1.2 Review on the mechanical behavior of pure and reinforced composites

Numerous studies have examined how different fiber characteristics, such as type, volume, angle, and configuration, affect the mechanical characteristics of fiber-reinforced 3D-printed polymer composites. Pure PLA is most ductile and has optimal tensile strength and fracture energy, whereas PLA reinforced with carbon fibers is the most brittle and has good bending strength during printing using the parameter extrusion temperature of 220 °C, layer height of 0.1 mm, infill density of 95%, linear infill pattern, nozzle diameter of 0.4 mm, and printing speed of 35 mm/s [129]. In addition, when comparing the tensile modulus EI of the reinforced PLA with 15% carbon fiber (CF) to the same property of PLA, the stiffness rose by 2.2%. The quasi-brittle matrix structure of PLA + CF presents the highest stiffness when compared with glass fiber of 200–300 µm, a carbon fiber of 100–300 µm, bronze particles of 5–30 µm with an amount in percentage of 15%, 10–15%, and 20%, respectively.

According to a study [130], the PLA + CF's Poisson coefficients ν_{12} and ν_{21} were far apart (ν_{12} was 2.5 times greater than ν_{21}). The materials chosen were PLA and PLA reinforced with short carbon fibers in a weight fraction of 15% (PLA + CF), having an estimated length of about 60 µm. The microstructure of the reinforced material was highly oriented when a rectilinear infill pattern and a volume fraction of 100% were applied. A tensile strength of 3596 and 7665 MPa at 0°, modules of 3340 and 4145 MPa at 90°, and shear modulus of 1140 and 1270 MPa at $\pm 45^\circ$ were found for PLA and PLA + CF, respectively.

A tensile modulus of 3.4 GPa and tensile strength of 369 MPa were found for a carbon fiber (45%) composite with 0° isotropic angle at three numbers of concentric rings with a nozzle diameter of 0.4 mm, raster width of 0.4 mm, raster angle of $\pm 45^\circ$, infill density of 100%, total shell thickness of 0.8 mm, and a flat build orientation. The result has superior performance to onyx, glass, and Kevlar fiber, demonstrating

the potential of carbon fibers in applications that demand higher tensile properties. The tensile properties are significantly improved with reinforcement with continuous fiber. The effect of reinforcement in carbon is more than glass and Kevlar in tensile strength, while in terms of tensile modulus, carbon is greater than Kevlar and Kevlar is greater than glass [131].

By using an infill density of 100%, travel speed of 50 mm/s, nozzle temperature of 240 °C, bed temperature of 80 °C, and layer thickness of 0.2 mm, it is found that 6% in weight tin nanoparticles blended to ABS increased flexural strength by 36.9%, flexural modulus of elasticity by 42.3%, and flexural toughness by 54%, while increasing tensile strength and stiffness by 18.1% and 22.2%, respectively, compared to pure ABS [132].

A pure ABS (20%) with recycled ABS (80%) produced Young's modulus of 2329 MPa, a yield strength of 34.81 MPa, and an ultimate tensile strength of 40.82 MPa [133]. The result is close to pure ABS, whereas strain hardening and material stiffening occur during material blending when the extrusion temperature is raised from 190 to 195 °C. The addition of abaca bracts biochar of spathe up to 2 wt% to ABS results in the highest fatigue strength and improves mechanical properties, while more biochar reduces the fatigue life. Because of the increased solid lubricity brought by the biocarbon addition, wear loss was decreased.

The investigation found that a blend of ABS combined with styrene ethylene butadiene styrene (SEBS) and ultra-high (UHMWPE) lowered the mechanical property anisotropy in terms of relative UTS to a difference of $22 \pm 2.07\%$ as compared to $47 \pm 7.23\%$ for samples printed from ABS with a nozzle diameter of 0.8 mm, a raster width of 0.6 mm, rectangular fill pattern, orientation angle of 0°, and infill density of 100% [134].

A FDM prints multi-material, ABS, PLA, and high-impact polystyrene (HIPS) using a printing parameters including nozzle diameter of 0.3 mm, sliced to form the four layers, layer height of 0.27 mm, printing speed of 30 mm/s, travel speed of 130 mm/s, extrusion temperature of 250 °C, bed temperature of 55 °C, rectilinear fill pattern, with an infill percentage of 60, 80, and 100% and printing speed of 50, 60, and 70 mm/s [135]. Young's modulus values of 325 MPa, 47.9 MPa, 175 MPa, and 112.5 MPa were found for the multi-material components, PLA, ABS, and HIPS, respectively, while flexural strengths of 2.96 MPa, 2.01 MPa, 9.07 MPa, and 7.04 MPa were found for the multi-material components, HIPS, PLA, and ABS, respectively.

According to [136], a maximal bending force of 68 N for aged PLA samples, which has an increase of 8.8% compared to the new PLA, and the aged PLA-X material had an increase of 7.7% in the bending force compared to the new PLA-X samples. These samples were printed using

concentric pattern, 30% infill density, layer height of 0.2 mm, two layers on the top, two on the bottom, and two from both sides, the nozzle temperature of 195.5 °C, nozzle diameter of 0.4 mm, printing speed of 60 mm/s, a chamber temperature of 33.2 °C, and the temperature of the platform of 48.1 °C. The PLA samples had higher maximal force values, 6.2% for the new samples and 7.3% for the aged samples.

The mechanical properties and impact strength are enhanced as the hemp hurd (HH) bio-composite particle size increases from 35 to 160 µm in PLA, while particle size above the printed layer thickness declines, which is investigated by [137], while the surface roughness does not alter when particle sizes change from 35 and 50 µm, which are lower than the layer height of 150 µm. In this investigation, a nozzle temperature of 200 °C, print bed temperature of 60 °C, nozzle diameter of 0.4 mm, infill density of 100%, layer height of 0.15 mm, printing speed of 60 mm/s, and a flat orientation were used.

3.1.3 Microstructural analysis: porosity, crystallinity, and surface roughness

Studies investigate the surface roughness, porosity, crystallinity, and surface morphology of 3D-printed parts for various applications. Heat treatment is an effective method to control and improve mechanical properties. A study [138] found an optimized result of the crystallinity (transforming δ form into α) and overcame the anisotropy using a 40 °C thermal treatment of PLA material. A study [139] also concluded that annealing above the cold crystallization temperature (T_{cc}) increases crystallinity, while it does not significantly impact the porosity. The same study [140] found annealing above the T_{cc} of 3D-printed PLA increases crystallinity while not significantly impacting the porosity, whereas annealing below T_{cc} increases tensile strength while not significantly impacting both porosity and crystallinity.

The PLA filament colors, including white, black, blue, grey, and natural, affect the tensile strength and crystallinity, whereas in the white filament, the printing temperature affects crystallinity as per the investigation [141]. The addition of acetone vapor and annealing in FDM-printed ABS decreased mechanical properties, while vapor smoothing with 20 ml of acetone for 15 min improved the surface roughness, and annealing at 100 °C for 60 min enhanced tensile properties [142]. Post-processing on ABS with acetone at a temperature of 65 °C found smoother surfaces and enhanced up to 20% from pure ABS [143].

In contrast, incorporating carbon fibers into PLA increases crystallinity while negatively affecting the microstructure, raising the melting temperature, and lowering the cold-crystallization temperature as per the study [144]. The

addition of carbon fibers into PLA acts as a nucleating agent, which increases crystallinity, raises the melting temperature while lowering the cold-crystallization temperature, and affects the microstructure [145]. The same study [146] concluded that the structure morphology and thermal stability were enhanced during the incorporation of rare earth oxides, Er_2O_3 and Gd_2O_3 , in the nano-powder sample into the ABS specimen in FDM of 3D printing, whereas radiation shielding did not significantly change. The results are appropriate for nuclear technologies and diagnostic energy range in medical purposes. Furthermore, a study [147] enhances the crystallization, thermal stability, and glass transition temperature of ABS printed with the Arburg-plastic-free forming technology. The study concluded that macro-voids between the printed layers and micro-voids within the droplets have confirmed the presence of porosity, which decreased impact and tensile strength.

3.1.4 Thermal characteristics

Researchers study the thermal characteristics, including energy absorption capacity, optical part propriety, thermal gravimetric analysis (TGA), thermal imaging, glass transition, and melting temperatures, which are shown in Fig. 8. A result from differential scanning calorimetry (DSC) showed that the level of crystallinity of the 3D-printed PLA specimens did not directly influence the mechanical properties, as per the study [148]. In addition, a Young's modulus of 1538.05 MPa for PLA and a Young's modulus of 2637.29

MPa for carbon fiber-reinforced PLA at 90 mm/s was found. Furthermore, the morphological and colorimetric prosperity of ABS changed with the addition of UV radiation and action of temperature, whereas PLA samples experienced fewer changes in their properties, which means PLA showed better resistance than ABS [149]. In addition to this, a study [150] found that the glass transition temperature for blends ranged between the values of ABS and recycled ABS from electronic waste (e-waste). The blending is promising and has proven to be feasible, making it possible to recycle plastics from e-waste and contribute to the preservation of the environment. The atmospheric pressure plasma treatment significantly improved the wettability of the polymer surface, with an increase to 60% for both working gases, which reduced its roughness by 50%, while the thermal conductivity was harmed by voids dispersed within 3D-printed ABS [151].

3.1.5 Other tests

Achieving an optimum dimensional geometry, weight, and printing time during the printing of material using FDM is another challenge. Thus, researchers investigate solving these parameters. The study [153] found a minimum buildup time of 9.830 min using an optimum parameter combination of speed of deposition of 40 mm/s, layer thickness of 0.4 mm, and infill density of 25% on FDM of ABS using Taguchi DOE. According to an investigation [154], the printed ABS specimen with a hollow internal pattern structure had

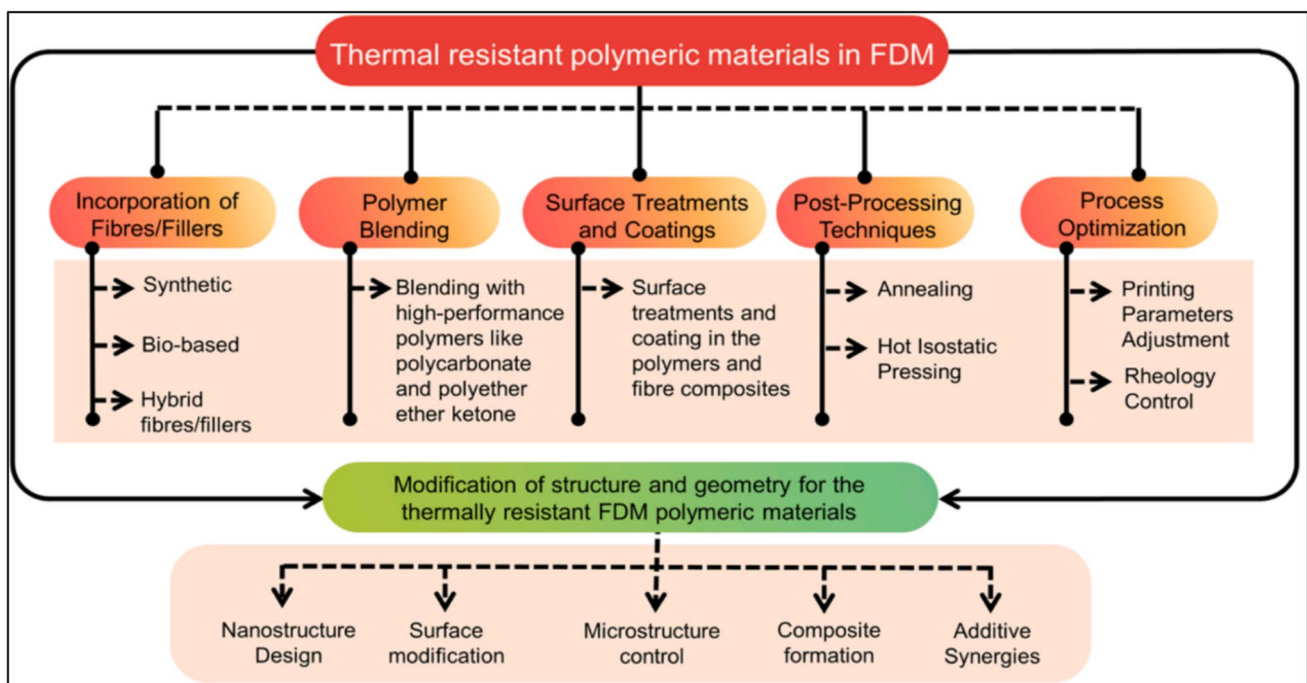


Fig. 8 The different strategies for improving the thermal properties of FDM-printed polymeric materials [152]

a dimensional accuracy of -0.19666 mm and surface roughness of $1.41 \mu\text{m}$, while the solid investment casting pattern PLA had a dimensional accuracy of -0.12777 mm and surface roughness of $3.07 \mu\text{m}$. For PLA and ABS, the average dimensional accuracy was 0.20401 mm and 0.21896 mm, respectively.

3.2 Fused deposition modelling process parameters and their effect

3.2.1 Printing speed

Printing speed, which ranges from 10 to 170 mm/s, is inter-related with the flow rate and directly affects the period of the build process and indirectly influences part strength and quality. Higher printing speeds reduce manufacturing time but often worsen geometrical tolerance and part strength due to voids and gaps in the layers, and it limits the solidification of bottom layers during deposition.

A study on ABS material [155] examined printing speeds of 10, 30, 50, and 70 mm/s, finding that decreasing printing speed improved surface smoothness, although surface roughness was less affected in thinner layers. von Mises stress during the cooling phase reached a maximum of 37.08 MPa, with deflection values of 0.91 mm, 0.93 mm, and 0.87 mm observed at speeds of 10 mm/s, 30 mm/s, and 70 mm/s, respectively, for layer deposit process where other printing parameters are constant. At $\pm 45^\circ$ printing direction maximum stress of 35.84 MPa and highest deformation of 6.5% is found.

3.2.2 Infill pattern

The infill pattern refers to the structure, shape, and technique used to fill the molten filament in the process of FDM. Commonly utilized infill patterns, shown in Fig. 9, significantly influence both the mechanical characteristics of printed parts and printing time. A study [156] evaluated linear, triangular, and concentric infill patterns with an infill density of 75, 80, and 85% and layer thickness of 100, 200, and 300 μm on ABS material. The result concluded that concentric patterns, along with 80% of infill density and 100 μm of layer thickness, resulted in 123% and 115% higher tensile strength and 168% and 80% higher impact strength compared to linear and triangular patterns, respectively, with the same printing parameters. The concentric pattern promotes interlayer bonding between layers and results in a tensile strength of 38.95 N/mm^2 .

In terms of energy absorption, honeycomb patterns with square and hexagonal unit cells demonstrated superior performance for blending ABS and TPU [157]. The square unit cell of the honeycomb pattern achieved energy absorption values of 2.2 kN.mm for TPU and 11.5 kN.mm for ABS, and hexagonal honeycombs resulted in energy absorption of 15.1 kN.mm when the combination of TPU-ABS is altered. For both square and hexagonal unit cells, a layer thickness of 0.2 mm, bed temperature of 85°C , infill of 100%, flow rate of 100%, with a printing speed of 55 mm/s for ABS and 25 mm/s for TPU, were utilized. The square honeycomb was 66% less than that of the hexagonal TPU specimen in energy absorption, which has a large difference in topology.

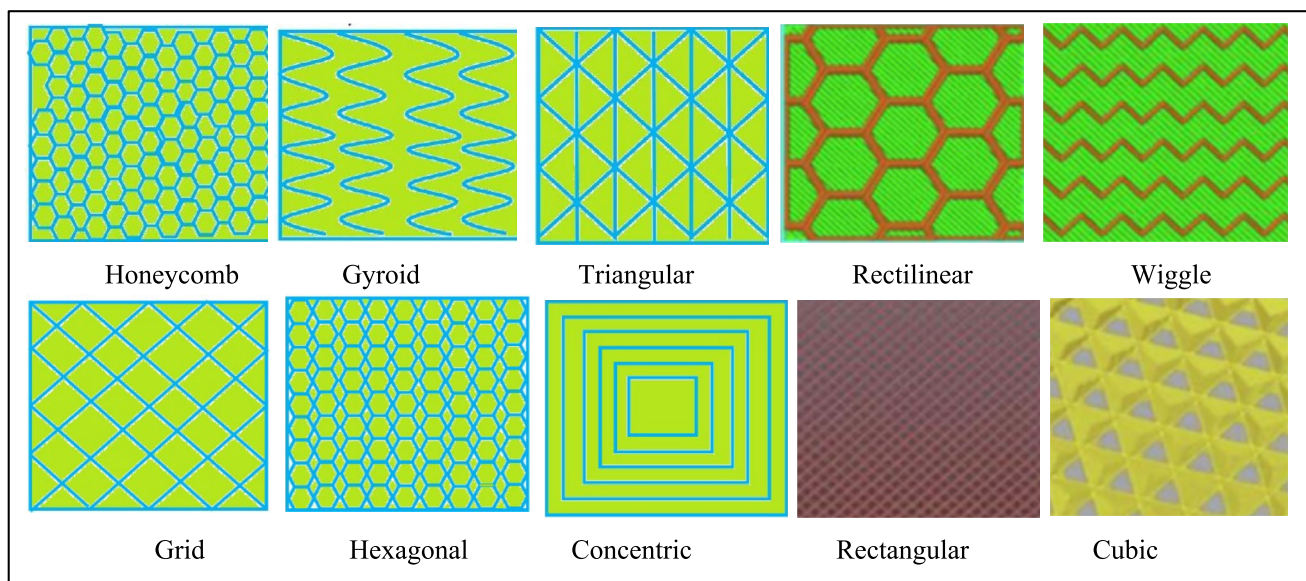


Fig. 9 The different infill patterns used in FDM of different materials [160]

To prevent stock accumulation in ship power supply insulators, ABS and PLA were printed in FDM at a constant printing speed of 70 mm/s, layer thickness of 0.35 mm, and layer height of 0.2 mm. The maximum relative dielectric constant of 2.9989 was found at maximum infill density (100%) with a triangular pattern, while the minimum value of 2.197 was found with square infill with a minimum infill density of 33% for ABS. In addition, the relative dielectric constant of 1.778 was found for PLA that is obtained at a corrugated infill pattern with an infill density of 33%, while the maximum value of 2.814 was found at an infill density of 100% for all infill patterns [14].

A continuous fiber-reinforced composite (KFRCP), onyx (combination of chopped carbon fiber with nylon) with a nominal diameter of 1.75 mm, and continuous Kevlar fiber at 0.35 mm in diameter was printed with the same printing parameter regardless of reinforcement and pattern. The result showed that the tensile strength of KFRCP rises 11 times and the impact strength increases by 3 times than virgin onyx. A minimum tensile strength of 21 MPa was found with zero Kevlar, while it increases 12.5 times for KFRCP at a reinforcing layer of 12. A maximum tensile strength of 34 MPa was found at continuous Kevlar fiber with 90° orientation angle and 10 layers of fiber. A tensile strength of 264 MPa was observed in Kevlar with 12 layers of fiber at isotropic zero pattern and 0° orientation, while concentric patterns produced a tensile strength of 183 MPa, and isotropic patterns reached 223 MPa [158].

The gyroid pattern with a 75% infill and 90° orientation angle demonstrated a compressive strength of 27 MPa and modulus of 658 MPa, with compressive properties increasing as infill density rose from 25 to 75% [159]. with increasing printing speed at a constant infill pattern, the compressive properties increase.

3.2.3 Layer thickness

Layer thickness (LT), also known as height, is influenced by several printing parameters, including raster width, nozzle tip size, printing speed, material type, and nozzle diameter. An optimal combination of printing parameters for enhancing the tensile strength of PLA includes a layer height of 0.2 mm, zero rotation angle, and a printing speed of 20–30 mm/s. In contrast, a 45° rotation angle tends to reduce tensile strength due to increased alignment of the fibers parallel to the loading direction. At a constant orientation, using a nozzle diameter of 0.4 mm, 100% infill density, a printing temperature of 210 °C, and a fixed printing speed, excessively low or high layer thickness can negatively affect the mechanical properties of the printed part [161].

A hardness of 79.6 HS (D) was achieved for both shell and infill configurations of ABS specimens manufactured at a 0.2 mm thickness [162]. A maximum of hardness value

is achieved at the lowest layer height of 0.1 mm with 100% infill in the shell and infill pattern, and decreases almost linearly for the 0.2 mm layer height at a linear infill pattern. In addition, heating the build plate to 100 °C at the same infill and layer height has a 1% reduction in hardness, which is a minor impact. For both infill patterns at a constant bed temperature of 100 °C, extruder temperature of 250 °C, nozzle diameter of 0.4 mm, and printing speed of 30 mm/s, as the layer height reduced from 2 to 1 mm, the hardness increases and the production time increases nonlinearly.

Further studies [163] agreed that a minimum layer thickness of 0.1778 mm resulted in a dense microstructure with a void density of 3.34% and higher diffusion between layers of 69.29%. The ABS have a tensile strength of 34.07MPa at that layer height with a 0/90 raster angle and flat build orientation. While increasing the layer thickness to 2 mm at a constant nozzle temperature of 210 °C, infill density of 10% during printing of PLA, the mechanical properties are decreased due to correlating with increased diffusion between layers. The layer height was reduced to 0.14 mm to get higher precision and a smoother surface [164]. The elastic modulus increases from 0.269 to 0.272 MPa and dynamic viscosity from 81.008 to 86.242 MPa.s by reducing the layer thickness from 0.33 to 0.254 mm, with the same printing parameter including print temperature of 300 °C and infill density. An equivalent modulus of 503.47 MPa found at 0° orientation, and 0.254 mm layer height, that is 6.7% higher than the result found at layer height of 0.33 mm and 45° orientation. In addition, equivalent moduli of 0.272 MPa and 0.233 MPa were found at 0° with 0.254 mm and 45° with 0.33 mm, respectively, for creep phenomena [165].

Different studies show that the layer thickness ranges from 0.06 to 0.6 mm, with most of the studies focusing on 0.10–0.30 mm. Thicker layers shorten production time but reduce part resolution, while thinner layers increase resolution at the expense of longer printing times.

3.2.4 Infill density

Infill density ranges from 20% for non-load-bearing parts to 100% for functional components [166]. At 50% of infill, Young's modulus of 342 MPa was found. While at 100% infill, the tensile strength increases by 51% than 50% infill at constant temperature. Reducing the infill to 25% resulted in a 12% decrease in the tensile strength compared to the 50% infill. Additionally, using a honeycomb infill pattern increases the printing time by 45%. The highest tensile strength of 34.02 MPa and Young's modulus of 476 MPa were achieved with 100% infill and a printing temperature of 230 °C using a honeycomb pattern [167]. The printing temperature has a higher effect at lower infill density.

In addition, at a constant infill density of 40%-layer thickness of 0.2 mm, nozzle temperature of 210 °C, print angle $\pm 45^\circ$, and a linear infill pattern, reducing the printing speed from 40 mm/s to 20 mm/s results in an increase in tensile strength from 20 to 22 MPa and an increase in weight from 6.7 to 6.9 g, while ductility is significantly affected. When the layer thickness is reduced from 0.2 to 0.1 mm while keeping other parameters constant, the tensile strength slightly decreases from 24.2 to 23.9 MPa, the weight increases from 8.39 to 8.78 g, and the printing time nearly doubles, increasing from 47 to 92 min. The study concluded that an infill density of 40% offers an optimal balance between material usage, cost, and time efficiency [168]. As infill increases from 20 to 80%, the strength and stiffness of the PLA increase, while it exhibits brittle behavior at constant parameters. In addition, infill density increases from 20 to 80%, and layer thickness decreases from 0.2 to 0.1 mm, the printing time increases from 31 to 91 min, and strength increases from 13.8 to 23.9 MPa and weight increases from 5.16 to 8.78 g.

Furthermore, the same study utilizes a constant nozzle temperature of 220 °C, bed temperature of 60 °C, nozzle diameter of 0.4 mm, layer thickness of 0.2 mm, printing

speed of 40 mm/s, line infill structure, and raster angle of $\pm 45^\circ$ by varying the infill density from 50 to 100% for PLA material [169]. The optimal printed quality values are found in the range of 70 to 100% infill and with higher densities improving strength but increasing brittleness and making it more brittle in PLA. High impact energy of 4.72 kJ/m², Izod strength of 4.2 kJ/m² were found at 100% infill and a tensile strength of 48.4 N/mm² and Izod of 1.7 kJ/m² found at 100–75–100% arrangement of infill density. In contrast, an impact energy of 1.9 kJ/m² was found at 75–50–75% arrangement of infill density. Figure 10 illustrates the infill density of different infill patterns.

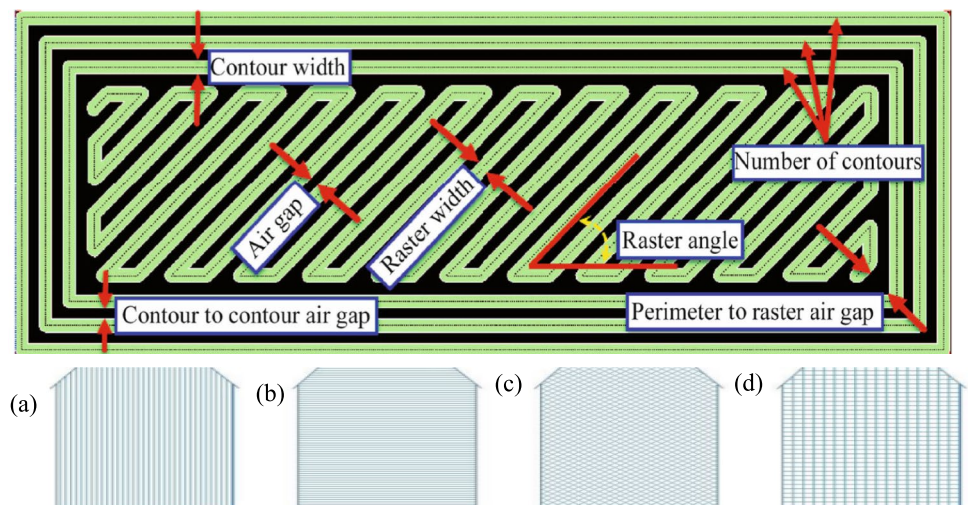
3.2.5 Raster angle

The arrangement of the successive lines of a layer in a 3D FDM-printed part refers to the raster angle that ranges from 0 to 90°, shown in Fig. 11. An alternating raster is represented by θ_1 , θ_2 , and θ_3 , while the unidirectional raster is known. According to the results in [171], when keeping the parameters constant such as a layer thickness of 0.15 mm, nozzle diameter of 0.14 mm, infill density of 100%, and plate temperature of 100 °C for ABS and 60 °C, for both

Fig. 10 The effect of the infill density on different mechanical properties [170]



Fig. 11 Raster angle orientation and paths: **a** 90, **b** 0, **c** 45, **d** 0 and 90 [175, 176]



PLA and PETG, as well as nozzle temperature of 220 °C for ABS, 190 °C for PLA, and 230 °C for PETG-PLA printed at a 45° angle exhibited approximately 3.25% higher tensile strength than at a 90° angle. In contrast, ABS and PETG samples printed at 0° and 90° angles showed 11.9% and 12.9% higher tensile strength, respectively, compared to those printed at a 45° angle. Additionally, in compression testing, the samples printed at a 45° angle showed approximately 3.15% higher strength for PLA and 2.35% for PETG, while ABS samples exhibited no significant difference.

The study [172] confirms energy absorption is higher at a 45° raster angle, offering superior elastic and plastic deformation characteristics. Furthermore, a tensile strength of 21.93 MPa was achieved at 0° and 12.13 MPa was obtained at 90° during the printing of ABS [173]. The same study gets an optimal tensile strength at raster angles between 23 and 31° relative to load direction. However, the raster angle does not affect the modulus of elasticity. These show parts printed perpendicular or transverse to the loading direction exhibit lower tensile strength compared to the parallel configuration, and the mechanical properties were higher for an alternating raster angle than a unidirectional raster angle [174].

3.2.6 Build orientation

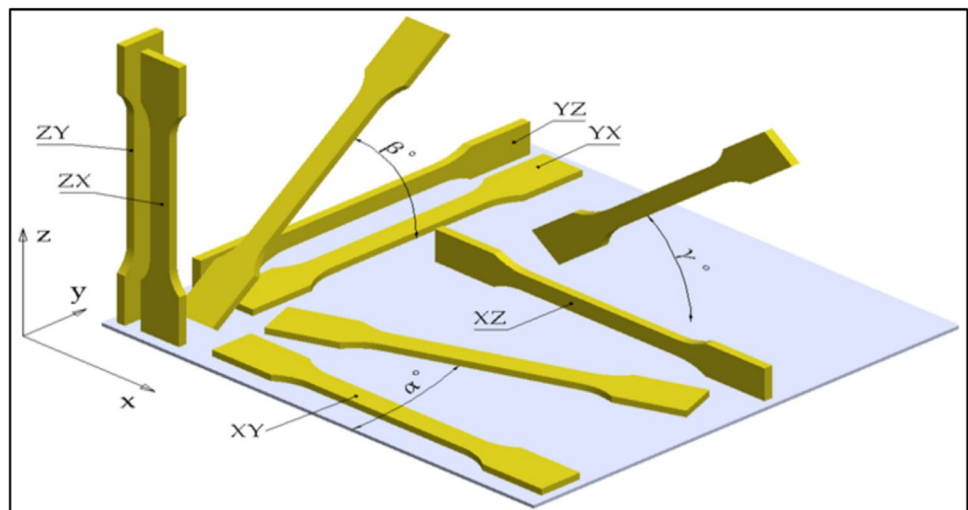
The alignment of the X, Y, or Z axes of the FDM machine within the build platform and the printing angle can be classified into three groups, namely flat orientations (XY and YX), on-edge orientations (XZ and YZ), and upright orientations (ZX and ZY), while other orientations are formed by rotating these configurations at specific angles [177] shown in Fig. 12. According to [178], under static loading conditions, the PLA sample exhibited a tensile strength of 38.7 MPa in the X build orientation which is 64% higher than that of built in the Y orientations (31.1 MPa) and a 45° orientation (33.6 MPa), while other printing parameters are

held constant including a layer thickness of 0.20 mm. Under cyclic loading, the 45° orientation shows superior fatigue life compared to the X and Y orientations for the same applied load. In the horizontal orientation, a tensile strength of 38.68 MPa for commercial, and 30.19 MPa for manufactured ABS at printing parameter of layer height of 0.15 mm, infill density of 100%, print speed of 60 mm/s, build temperature of 60 °C, nozzle temperature of 200 °C, and triangular infill pattern were found. In the vertical orientation at the same parameter setting, a tensile strength of 53.84 MPa for commercial ABS and 31.52 MPa for manufactured ABS was found. In the inclined orientation, a tensile strength of 51.71 MPa for commercial ABS and 42.41 MPa for manufactured ABS was found [179].

The Young's modulus of horizontally oriented specimen exhibits 350 to 385 MPa and a flexural modulus of 4.8 to 5.3 GPa, whereas in vertically printed parts, 275 to 325 MPa and 3.6 to 4.4 GPa, respectively, were recorded, during the printing of ABS in FDM at constant nozzle temperature 210 °C, platform temperature 70 °C, infill density 100%, and layer thickness 0.178 mm [180]. A 15 wt% of short carbon fiber (CF) reinforced polyamide (PA) composite has resulted in a tensile strength of 96 MPa and a Young's modulus of 7.9 GPa, with an increment of 34 and 47%, respectively, when compared to the neat PA, in the XY build orientation. In addition, it exhibited a tensile strength of 96 MPa and a Young's modulus of 7.9 GPa in the XY orientation with constant nozzle diameter of 0.40 mm, nozzle temperature of 270 °C, bed temperature of 110 °C, infill density of 100%, raster angle of $\pm 45^\circ$, layer height of 0.20 mm, and infill speed of 40 mm/s. However, the fracture roughness of the PA/CF composite 3D printed part shows minimal enhancement from pure PLA [181].

A study by [182] reported a tensile strength of 30.60 MPa and a maximum load of 551.13 N in the axial direction with a 0.3 mm layer thickness for 3D-printed ABS specimens.

Fig. 12 Various build orientations of a part [185]



As revealed in the study, research has consistently found higher mechanical properties in the X and Y directions compared to the vertical Z direction. In addition, due to inter-layer breakage, the ZX and ZY orientations exhibit significantly lower mechanical properties than the XY, YX, XZ, and YZ orientations. Moreover, as the build orientation changes from 0 to 90°, mechanical properties tend to decrease, while the cyclic tensile load at a 45° orientation demonstrates a longer fatigue life than at the 0° orientation.

Further, a study [183] found a tensile strength of 23.75 MPa, 52.4 MPa, and 57.58 MPa at a building orientation of vertical, on side and horizontal respectively, at a bed temperature of 45 °C, printing speed of 60 mm/s, extrusion temperature of 210 °C, layer thickness of 0.2 mm, nozzle diameter of 0.4 mm, cooling fan speed of 100%, and infill density of 100% for a PLA material fabricated in FDM.

The studies [184] confirm a horizontal (ZY) orientation found a maximum tensile strength of 31.41 MPa, which is 25% greater strength compared to vertically (YZ) oriented printed specimens with constant infill density of 100%, printing pattern of (grid and zigzag), print speed of 30 mm/s, bed temperature of 100 °C, extrusion temperature of 235 °C, and layer height of 0.2 mm. The printing orientation has a greater impact on mechanical behavior compared to the number of shells or infill patterns.

3.2.7 Nozzle diameter

Nozzle diameter plays a critical role in filament deposition and the mechanical properties of 3D-printed parts. The study [186] found that increasing the nozzle diameter from 0.3 mm to 0.6 mm reduced voids between adjacent strands, which enhanced the density and the ultimate tensile stress (UTS) of PLA materials. The thickness of each layer was set at the ratio of 20% of the nozzle hole diameter. A maximum density of 1.214 g/mm³ and tensile strength of 45.1 MPa obtained at 0.6 mm with a constant bed temperature of 60 °C, infill density of 100%, extruder temperature of 200 °C, line infill pattern, and printing speed of 80 mm/s. At a constant layer height of 0.2 mm, 100% extrusion ratio, nozzle temperature of 225 °C, and a bed temperature of 60 °C, tensile strengths of 30.02 N/mm² and 29.61 N/mm² were found at nozzle diameters of 0.4 mm and 1 mm, respectively, with a printing angle of 0°. Tensile strengths of 30.35 N/mm² and 19.68 N/mm² with a printing angle of 45° and a tensile strength of 17.07 N/mm² and 11.89 N/mm² with a printing angle of 90° were found at nozzle diameters of 0.4 mm and 1 mm, respectively. Optimum mechanical properties of carbon black/PLA in terms of stiffness and strength were obtained for the configuration of printing at 0° direction with the largest nozzle (1 mm diameter) and the smallest layer height (0.05 mm) [187].

As nozzle diameter increases from 0.4 to 0.8 mm, infill density from 95 to 105%, and raster angle from ±45 to 90°, the tensile strength increases from 38 to 42 N/mm², elastic modulus increases from 710 to 725 MPa, while the layer thickness reduces from 0.17 to 0.06 mm. These results were found at a constant platform temperature of 90 °C, extrusion temperature of 250 °C, printing speed of 30 mm/s, and layer thickness of 0.1 mm. Similarly, road widths of 0.48 mm and 0.96 mm have been shown to improve mechanical properties. This is due to gravitational forces effectively spreading the melted plastic at larger diameters [188]. A tensile strength of 46.07 N/mm² was found with a nozzle diameter of 0.6 mm, liquefier temperature of 230 °C, extrusion velocity of 30 mm/s, filling velocity of 20 mm/s, layer thickness of 0.3 mm, no air gap, and with 5 number of contours [189]. While a tensile strength of 31.628 N/mm² and building time of 6.23 h was found with a nozzle diameter of 0.2 mm, liquefier temperature of 200 °C, extrusion velocity of 20 mm/s, filling velocity of 20 mm/s, and layer thickness of 0.1 mm. The minimum building time was 0.52 h, which is found at a nozzle diameter of 0.6 mm, liquefier temperature of 215 °C, extrusion velocity of 25 mm/s, filling velocity of 30 mm/s, and layer thickness of 0.3 mm. Among these nozzle diameters, ranging from 0.5 to 1.5 mm contributed the least to UTS during ABS printing.

3.2.8 Nozzle temperature

Nozzle temperature is another critical parameter in FDM, which depends on the melting point of printed material, which is the most studied process parameter. The study [190] found a maximum tensile strength of 50.16 MPa and an elastic modulus of 4340.38 MPa at a printing temperature of 230 °C during printing of PLA in the FDM method. At the printing speed of 60 mm/min, the tensile strength and elastic modulus are 51.47 MPa and 5102.12 MPa, respectively. When the printing temperature is between 200 and 230 °C, the tensile strength has increased by 4.3%, elastic modulus has increased by 11.06%. The printing temperature is between 230 and 240 °C, and the tensile strength and elastic modulus have decreased by 1.02% and 3.32%, respectively. When the printing speed is 60 mm/min, a maximum tensile strength and a maximum elastic modulus of 51.47 MPa and 5102.12 MPa, with an average growth rate of 2.34% and 10.33%, respectively, found at a constant nozzle diameter of 0.4 mm.

As printing temperature increases from 210 to 230 °C, the porosity content decreases from 6.94 to 5.77%, while the fracture toughness increases from 2.68 to 3.56 MPa. The results were found at constant process conditions of nozzle diameter of 0.4 mm, layups of ±45°, layer thickness of 0.2 mm, infill density of 100%, and vertical speed of 150 mm/s. The stiffness and toughness of the 3D-printed

PLA are balanced by the linear decrease in the elongation at break when the printing temperature increases from 170 to 230 °C [191].

Similarly, in [192], a bed temperature of 60 °C and a nozzle temperature between 190 and 210 °C with a printing speed of 40 to 80 mm/min give higher tensile properties and lower roughness values for PLA polymer fabricated with FDM. It was also a sufficient geometrical accuracy of PLA parts was found at nozzle temperatures in the range of 170–190 °C with 40 mm/min over the other parameter combinations. A nozzle temperature of 190 °C with a printing speed of 40 mm/min found a good surface finish, while for process productivity, a printing speed of 80 mm/min and 190 °C temperature was considered. In addition, a 190 °C temperature and 40 mm/min found a maximum elastic modulus than 80 mm/min. Increasing nozzle temperature initially improves tensile strength and elastic modulus while causing degradation at excessive ranges. Numerous studies agree the optimal nozzle temperature ranges between 190 and 230 °C during biodegradable material printing.

3.2.9 Extrusion temperature

Extrusion temperature ranges from 165 to 185 °C and screw rate of 2–6 rpm for a 3D850D grade PLA manufactured using 3D FDM, yielding a tensile strength 18.6–27.9 MPa at a raster angle of 90°, bead temperature 60 °C, printing speed 40 mm/s, nozzle diameter 0.4 mm, layer height 0.2 mm, line pattern, and infill density of 100% [193]. However, with the same printing parameter, a combination of extrusion temperature from 175 °C and extrusion speed of 5 rpm yields maximum tensile strength of 27.9 MPa.

An optimum deformation was found during the printing of ABS using an optimal extrusion temperature of 175 °C. Beyond 250 °C, the residual stresses and deformation increased at a chamber temperature of 25 °C, extrusion speed of 60 mm/s, infill density of 10%, and plate temperature of 55 °C [194]. Chamber temperature has a strong effect and extrusion temperature has a significant influence on the mechanical behavior of the part, while extrusion speed has a noticeable impact on the material extrusion process. Minimum residual stress and part deformation are found in the extrusion speed of 50 to 70 mm/s. To minimize this effect, the chamber temperature could be maintained.

The same study [195] found a tensile strength of 32 MPa at an extrusion temperature of 210 °C, a nozzle diameter of 0.3 mm, and a tensile strength of 50 MPa at an extrusion temperature of 240 °C and a nozzle diameter of 0.5 mm. An infill density of 20% was unacceptable, while 80% and 100% have no difference in strength, while only time consumption and waste material were highlighted. Other printer parameters remained constant throughout the study. The results

show that two printing parameters, the temperature of the extruder and the nozzle diameter, were more significant for the mechanical properties. The bed temperature of 60 °C, and the printing temperature of 60 mm/s.

Furthermore, a compressive strength of 41 N/mm² for linear infill with 45° angle and 35 N/mm² for honeycomb infill, while a compressive modulus of 918 N/mm² and 868 N/mm² for linear infill with 45° angle and honeycomb, respectively [196]. These results were found for both infill with a layer thickness of 0.1 mm and infill density of 40%, while extrusion temperature of 209 °C to 254 °C for linear infill with 45° and extrusion temperature of 255 °C for honeycomb infill during the printing of ABS as per the study. The results show that the infill density has the highest influence on the printing properties, followed by the layer thickness and, finally, the extrusion temperature as the lowest influential factor.

Based on the above works of literature, too low a temperature will not melt the material, resulting in clogging of the nozzle, while excessive temperature will result in void generation, thereby impacting the final strength of the product. Therefore, an optimum temperature is required for a better-quality product.

3.2.10 Build plate temperature

The temperature of the 3D printer bed significantly influences the adhesion between the first printed layer and the printing bed. Proper adhesion is essential to prevent part warping and to enhance dimensional accuracy.

A study [197] uses constant printing parameters including nozzle temperature of 210 °C, bed temperature of 60 °C, raster angle of $\pm 45^\circ$, printing speed of 50 mm/s, travel speed of 200 mm/s, and infill density of 100% during printing of PLA increased impact strength from 35 to 63 J/m, while the heat deflection temperature (HDT) rose from 55 to 75 °C when the bed temperature increases from 60 to 105 °C. A bed temperature increased from 45 to 105 °C, the crystallinity increases, and as bed temperature rises from 60 to 105 °C, the mechanical properties including in strength and modulus were improved. Another research [198] showed that a maximum temperature value was observed at places near the corners of the plate structures, with a bed temperature of 75 °C with a constant printing parameters of including building chambers temperature of 75 °C, extrusion temperature of 270 °C, raster orientation of $\pm 45^\circ$, a layer thickness of 0.245 mm, and 13 layers (3.302 mm height).

As per the study of [199], aluminum plates were discovered the highest tensile strength, stainless steel, and nimonic bed plates improved UTS by raising the printing speed 110 mm/s with constant printing parameters of

extrusion temperature of 215 °C, using aluminum plate, found best result, at constant infill density of 100%, layer height of 0.2 mm, line pattern, number of shells 2, and orientation of $\pm 45^\circ$. In addition, maximum specific energy consumption was obtained at speed of 110 mm/s and temperature of 215 °C using aluminum plate, while the worst results found for steel plate using speed of 70 mm/s at extrusion temperature of 215 °C.

To improve the quality of the printed material additionally heated the plate have an impact on the air temperature above the plate [200]. In contrast to open machines, the addition of built chambers can increase the temperature of the far environment inside the chamber, resulting in a cooler portion. In addition, build temperature play bed adhesion purpose and heats the air above it. Printing at low layer thicknesses, such as 300 μm , and adding enclosure improve temperature inside the chamber. Proper adhesion is essential to prevent part warping and to enhance dimensional accuracy. It has been reported that setting the bed temperature slightly above the glass transition temperature of the printing material provides optimal adhesion properties.

4 The design of experiment, parameter modelling, optimization technique, and numerical studies used in FDM

4.1 Design of experiments

Conventional techniques like the trial-and-error method are employed in technical and scientific research for the design of experimental runs. Currently, engineers and researchers in various disciplines utilize the design of experiments (DOE) to determine the process output, affecting factors, and analysis of experimental studies. DOE is a cost-effective and efficient statistical analysis tool for numerical simulation and experimentation [201, 202]. Full factorial, fractional factorial design (FFD), the Taguchi method, and response surface methodology (RSM) are the most commonly utilized DOE methods [203, 204]. Full factorial experiments are created with a combination of input factors (n) and every discrete level (k), while fractional factorials are utilized with a reduced number of experiments and costs. The full factorial design of experiments necessitates conducting numerous trial runs, which are costly, time-consuming, and too complicated, as per the study [205]. The RSM method of central composite design and Box-Benken techniques are also used by [206–208]. In addition, the Taguchi method minimizes the number of experimental runs, which reduces both time and cost.

An optimal process parameter of a layer height of 0.1 mm, an infill density of 100%, a printing velocity of 20 mm/s, and an extrusion temperature of 190 °C were found

using Taguchi orthogonal array DOE as per the study [209]. The study found a tensile strength of 10.15 MPa and an elastic modulus of 198.57 MPa for the FDM of PLA-based wood fiber composite. In addition, a study [210] uses the Taguchi orthogonal array method for printing a double layer of ABS and PETG, finding a tensile strength of 44.73 MPa and Young's modulus of 758.12 MPa at a printing speed of 30 mm/s, a layer thickness of 0.1 mm, and an infill density of 75%. Furthermore, a study [211] utilized Taguchi DOE for FDM on ABS specimens and found a tensile strength of 44.24 MPa experimentally and 51.58 MPa in a confirmation test.

4.2 Parameters modelling technique used in FDM

The focus of numerous researchers was on controlling the printing process parameters of FDM to enhance the quality and quantity of printed specimens to meet customer demands. Mathematical modelling is one of the statistical modelling tools used to predict, modelling, inline manufacturing, and confirm the accuracy of the experimental work. Different modelling methods, including regression model, ANN, fuzzy logic, and ANFIS model in printing parameters and outputs of FDM were discussed in this sub-section.

4.2.1 Response surface methodology modelling

The RSM model is considered to be fit well, if the value ($p < 0.01$) of the model is significant and the lack of fit is non-significant and further, the coefficient of determination (R^2) should be close to 1 [212, 213].

The regression model shows the relationship between the printer parameters and response properties [214]. The study found predictive models using the RSM method for compressive properties of ABS on FDM; the results are in line with the results of experimentation with an acceptable percentage error of 1.36% [215]. The predicted result using RSM was good within the experimental result of the tensile strength of FDM printed ABS [216]. In addition, RSM found a tensile strength of 50.54 MPa, showing good agreement with experimental results of FDM of PLA material at process parameters of layer thickness of 0.23 mm, nozzle temperature of 208 °C, and printing speed of 55.5 mm/s [217].

4.2.2 Artificial neural network modelling

The effect of the printing parameters and experimental results are predicted by artificial neural network (ANN) by considering the fitness function [218]. The ANN structure consists of three key players: inputs, hidden, and outputs, as shown in Fig. 13.

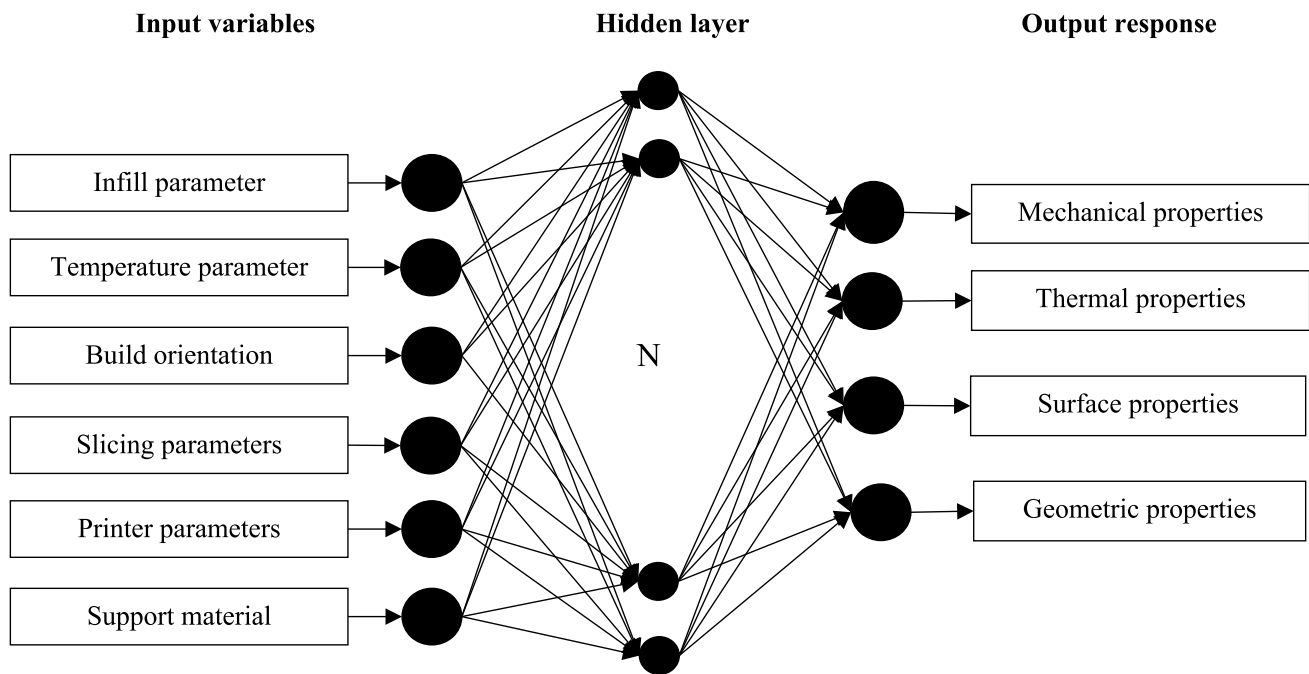


Fig. 13 Artificial neural network architecture with a single hidden layer [228]

The materials' properties are enhanced by developing composite materials, while FDM cannot handle complex geometries with ease, which opens up topological optimization and lightweight structural design. These topology optimizations can produce optimized geometries with net shape and strength-to-weight ratio [219]. In addition, high costs of production and expensive raw materials, approaches, including making models smaller and lighter, are especially considered in the design of structures. Topology optimization reduces materials consumption, times, and costs for producing layered manufactured products to the optimal shapes aligned with the objectives set early on [220].

The RSM, ANN, and simulation results strongly correlate with a maximum deformation of 0.101 mm and a von Mises stress of 29.1 MPa. A minimum safety factor of 1.2, a minimum thickness of 3.85 mm, and a maximum thickness of 8.42 mm are identified for optimal topology optimization using genetic algorithms (GA) for tough PLA in FDM. The optimal printing orientation is vertical orientation with greater hardness and wear resistance [221].

The same study found von-Mises stress of 24.65 MPa for the initial dense beams, 25.18 MPa for the topology optimization beam before, and 29.2 MPa for the topology optimized after applying cellular structuring using Abacus software on ABS. Overall, the study minimize the total structural volume of the beam structure under static loads and constraints with the remaining volume of the part as a design variable [222].

The volume was reduced in the final model by 66.29%, 52.88%, 50.05%, 51.85%, and 35.02%, for cantilever beam, double-girder beam, dome-shaped geometry, thermal loading, and hot fluid transfer tee, respectively, using ANSYS software. While the change in volume causes some increase in the tension, and displacement of the final model. The increase of stress in the cantilever beam, double-girder beam, and dome-shaped geometry was 88, 800, and 6 MPa, while the deformation in these sample increased by 10.2%, 200%, and 3.3%, respectively [223].

The study found maximum tensile strength of 0.56 KN, compressive strength of 60.58 MPa, impact strength of 4.84 J/cm², and hardness of 80 shore D in the octet and cubic infill patterns. The drone frame was modelled using Fusion 360 software with a stand load-carrying capacity with minimum material of PLA-20%Zn filament, and found that a cubic infill pattern gives a low-cost replacement for the currently used conventional frame [224].

A volume of 38.37 cm³, 26.17 cm³, and 26.173 cm³ was found before optimization, optimization along Y, and optimization along Z. While a mass of 0.04 kg, 0.0272 kg, and 0.027 kg was found before optimization, optimization along Y, and optimization along Z respectively during production of garage door control system (handle) with ABS, which manufactured by the FDM, ANSYS [225].

The weight of the optimized ABS body is 19.8% less than the original when reducing the stress in the most stressed parts of the volume. Topological optimization leads to a

reduction in the required volume and a related reduction in the stressed cross-sections in the entire body, which leads to a more significant overall deformation of the loaded part by 15% [226].

The prediction correlation coefficient (R) value for RSM was 80.63, and for ANN it was 99.12 for UTS for FDM of ABS material [227]. The R value obtained in ANN is 99.44, which is higher than the modelling obtained in RSM 97.12, which ANN is more reliable and accurate (90%) in predicting the results for ultimate tensile strength (UTS) and elastic modulus of ABS fabricated by 3D printing. While both models effectively analyze the effect of parameters on the mechanical properties. Both ANN and RSM modelling show that the number has the most effect than layer thickness and infill density on the UTS, and is proportional to each other.

A study found that the highest ANN performance was achieved with two hidden layers containing 150 neurons, resulting in an R^2 value of 0.87, which is close to 0.9,

indicating that the ANN model is highly effective and suitable for predicting the surface roughness of FDM ABS parts. Compared to experimental results, the model finds a decrease in the surface roughness with the increase in infill density to 100%, while layer height increases to 0.4 mm, print speed from 20 to 80 mm/s, and nozzle temperature from 190 to 230 °C, the surface roughness increases [229]. To enhance network performance, Trainlm is widely used as the training function, whereas Tansig, logistic-sigmoid, and purelin are commonly used as activation functions. The detailed process of the ANN and RSM models is depicted in Fig. 14.

A study [235] examines the tensile and compressive strength of different materials processed by FDM. The invertible neural networks (INN) mapping from the parameters to the output responses is a forward and inverse process. Seventy percent of the total data is employed for the training, 15% for the test, and 15% for validation, while the

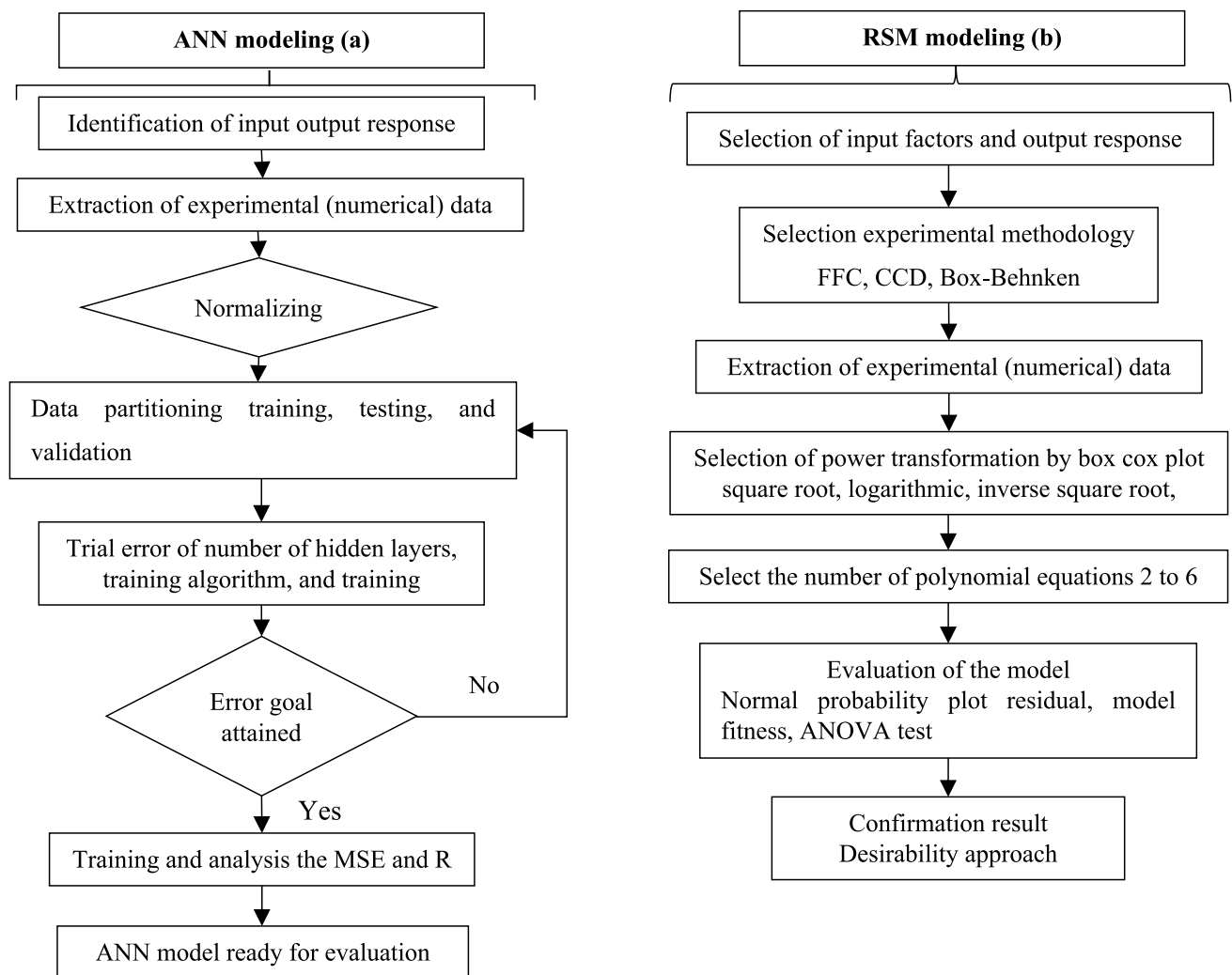


Fig. 14 Modelling procedure: **a** Steps execution of ANN implementation [230, 231] and **b** steps execution of RSM [232–234]

input data and outputs are specified and normalized between 0.1 and 0.9. The data result in good accuracy in terms of parts manufacturing with input accuracy of 85.34%, and in most cases, it rises to 99.96%.

4.2.3 Fuzzy logic and neuro-fuzzy inference modelling method

Unlike ANN, fuzzy logic (FL) does not require training data, while it demands a comprehensive set of rules, necessitating significant data storage and a database [236]. Regardless of predicting the tensile strength, the print speed and extruder temperature have less effect, while infill density has a larger effect on the tensile strength of PLA fabricated using FDM. The prediction model and experimental result are in line with a percentage error of 0.40% and 1.84% for ANN and FL, respectively [237]. Adaptive neuro-fuzzy inference system (ANFIS) exhibited strong predictive capabilities for the tensile strength of PLA, PLA-CF, and PC-ABS, with root mean square error (RMSE) values of 1.012, 0.5, and 0.835, respectively [238]. The coefficients of determination (R^2) for the predicted model were 0.98 for PLA, 0.99 for PLA-CF, and 0.89 for PC-ABS.

ANFIS has been widely recognized for its efficiency and effectiveness in predicting the response properties during the FDM process, on the integrated strengths of neural networks and fuzzy logic. The correlations between the process parameters and response variables are used to analyze the response and the plotted graph is used to analyze the effect of FDM parameters on the response. An experimental investigation [239] on the weight gain of PLA processed via FDM found that the ANFIS model produced results nearly identical to the experimental values, with an average percentage error of less than 5.45×10^6 . The ANFIS can effectively predict the mechanical properties of FFF on printed PLA parts, ultimately enhancing the efficiency and accuracy of the printed parts. A good correlation with the experimental result was found using ANFIS of the mechanical properties with an RMSE of 20% and 9.75% for stress and strain, while due to some deviation a 40% of Young's modulus was found [240].

The 3–3–3–3 structure of the ANFIS model yields the best prediction result and faster prediction performance with minimum error for part build time, surface roughness, and compressive strength of PLA. The ANFIS plot results in better surface quality found at lower layer thickness (0.1 mm), while higher infill density (70%) and raster angle (60°), infill structure has a negligible effect. In case of building time, higher layer thickness (0.3 mm), lower infill density (30%), and raster angle recommended (0°), while triangular and honeycomb patterns take more time than the other infill type. The best compressive strength

is found at higher layer thickness and infill density, while raster angle negatively affects it. The honeycomb and triangular infill structure gives higher compressive strength than linear and zigzag infill patterns [241].

4.3 Research trends on parameter optimization techniques in FDM

The complexity of process parameters in FDM makes it challenging to achieve the intended result. Therefore, determining the optimal process parameter value to improve the quality objectives is known as optimization [242]. Numerous researchers embrace important practical implications, such as improving mechanical characteristics, performance, surface finish, sustainability, and innovation, by fine-tuning the process parameters of the FDM 3D-printed items. To maximize these process attributes, different researchers employ various techniques and resources. This subsection covers research on various process parameter optimization techniques used in FDM.

4.3.1 Taguchi optimization method

The Taguchi method has been widely employed for single-response optimization. For instance, a study [243] utilized the Taguchi technique for single response optimization and found an optimum temperature of 210 °C, an infill angle of 45°, an infill density of 100%, and a layer thickness of 0.1 mm for maximum tensile strength during FDM printing on polyurethane polymer. In addition, a temperature of 210 °C, an infill angle of 45°, an infill density of 15%, and a layer thickness of 0.1 mm were optimum parameters to achieve the highest elongation at break. Another study [244] found that an optimal process parameter of a layer thickness of 0.15 mm, an infill density of 55%, and a nozzle temperature of 210 °C yielded improved surface roughness utilizing Taguchi optimization on signal to noise for PLA FDM. A similar study [245] investigates a higher tensile strength using the Taguchi optimization technique for a bi-layered composite of PLA-ABS and yields optimum parameters of a 75% infill density, a printing speed of 20 mm/s, 30 layers, and a bed temperature of 100 °C.

4.3.2 Grey relational analysis (GRA) and multi-response optimization

Grey relational analysis (GRA) combined with the Taguchi method has been extensively utilized in optimizing FDM process parameters for various materials. A study [246] identified an optimal parameter of FDM, a printing speed of 70 mm/s, a layer height of 0.1 mm, and a printing temperature of 220 °C, achieving a balance of good dimensional

accuracy and a low void presence of 1.673% using Taguchi-grey relational analysis (GRA) in conjunction with the Taguchi during FDM of PLA wood composite. The study also found a grey relational grade (GRG) of 0.782, which is closely aligned with the predicted value of 0.754. In addition, a study [247] uses Taguchi single-objective optimization and found a tensile strength of 48.10 MPa, production time of 16 min, and material consumption of 2.43 g during the FDM process of composite PLA materials. Multi-objective optimization resulted in an 8.31% improvement in the GRG. The same study [248] investigates the tensile strength and hardness of PLA and yields optimal parameters as a printing speed of 70 mm/s, a layer height of 0.3 mm, and a 40% infill density utilizing Taguchi-based GRA (TGRA).

In a different study [249], GRA was used to optimize the tribological characteristics (wear rate and friction coefficient) of FDM-printed materials. The optimal printing parameters were as follows: 0.1 mm, 230 °C, 0.28 mm, and 75 mm/s for ABS polymer; a layer thickness of 0.14 mm, a nozzle temperature of 220 °C, a line width of 0.31 mm, and a printing speed of 50 mm/s for PLA; the study used RSM and discovered that the wear rate increased linearly with the printing speed, but first decreased and then increased with layer thickness and nozzle temperature, before increasing. Additionally, the lowest coefficient of friction and linear wear was achieved with an optimum process value of 50% infill density and a layer thickness of 0.1 mm of FDM on ABS material, while 50% infill percentage and a layer thickness of 0.15 mm for PLA utilizing GRA [250]. The study advised against using a 75% or 100% infill density with a layer thickness of 0.2 mm for ABS parts, as they consistently exhibited higher surface roughness. For PLA-ABS blends, optimal tensile strength was achieved using GRA with validated parameters [251]. Furthermore, a study validated that GRA, combined with a full factorial design and ANOVA, consistently minimized friction and linear wear for both ABS and PLA materials [252].

4.3.3 Response surface methodology (RSM) and advanced techniques

The study [253] utilized multiple linear regression with the Taguchi method and concluded the fatigue life of nylon and ABS materials increased as the nozzle diameter decreased, while the effect was statistically significant for ABS material. The highest fatigue life was observed in the tri-hexagon structure. The simulation results show a 14% deviation from experimental values. Increasing the infill density and setting the extrusion temperature to 220 °C resulted in a maximum failure load of 1218 N as per the study [254]. The percentage error between response surface method (RSM) predictions and actual results was less than 5%. Using RSM-ANN, the study [255] found that at an extrusion temperature of 210

°C, a layer thickness of 140 microns, a bed temperature of 80 °C, and a print speed of 35 mm/s, a tensile strength of 9.8 MPa, an elastic modulus of 506.9 MPa, and an elongation at break of 1.9% were obtained. With a layer thickness of 150 microns, an extrusion temperature of 217 °C, a bed temperature of 88 °C, and a print speed of 35 mm/s, the study also discovered a Pareto optimal solution with a tensile strength of 18 MPa, an elastic modulus of 508 MPa, and an elongation at break of 2% using ANN to GA. Experimental results agreed with the ANN-RSM technique approach. A further investigation [256] produced a prediction error of 0.25% and a goodness function of 1.981076 for RSM-GA and 1.9964 for ANN-GA. The study found that the wear rate was reduced when the percentage of iron reinforcement in the ABS composite was increased.

A tensile strength of 51.03 MPa using GA-ANN and 48.35 MPa using GA-RSM was observed in the study by [257], using FDM on ABS material. The tensile strength improved by 38.4% when 49.77% weight polycarbonate was added to ABS. This rise was ascribed to well-bonded structures, sufficient flow characteristics, and partial miscibility. Maximum tensile strength and toughness were found at optimal parameters of infill density of 60%, extrusion temperature of 200 °C, raster angle of 45°, and layer thickness of 0.2 mm utilizing the NSGA-II algorithm optimization on FDM for PLA material as per the study [258]. In addition, a study [259] utilized an ANN with GA during FDM of ABS parts, achieving a maximum flexural strength of 2.67 GPa at an optimum nozzle diameter of 0.6 mm, layer height of 0.1 mm, fill density of 64.63%, printing speed of 29.54 mm/s, and raster orientation of 85.23°. The study also yielded significance values of 0.17, 0.15, and 0.51 for linear, rectilinear, and honeycomb infill patterns, respectively, using a regression model. The study [260] concluded that the Taguchi parameter design yielded accurate results with minimal experimental runs compared to other design experiment methods, while a hybrid approach combining RSM and GA was proposed to solve multi-objective optimization problems. A hybrid approach combining RSM and GA was proposed to solve multi-objective optimization problems. Furthermore, to enhance the quality and mechanical properties of ABS, PETG, and multi-material blends (50% of ABS and 50% of PETG), GA-ANN hybrids were utilized and achieved comparable results as per the study [261].

4.3.4 Machine learning and hybrid techniques

Machine learning (ML) methods, including ANN, K-nearest neighbor, ridge regression, and LASSO regression, have been employed to predict the dynamic response of aluminum and ABS materials. For ABS, natural frequency was the most significant predictor, while for aluminum, temperature and amplitude were critical [262]. In addition, a hybrid

approach coupling GA with four prominent machine learning algorithms, namely decision tree (DT), random forest (RF), ANN, and gradient boosting regressor (GBR), was revealed by [263] to enhance the accuracy and efficiency of surface roughness prediction for FDM-printed PLA material. The GA-DT model excelled, achieving an R^2 value of 0.9378 and predicting surface roughness of 5.636 μm , closely matching the experimental result of 6.127 μm . The application of ML gradient boosting regression demonstrated the advantages of 3D printing over traditional methods, including rapid turnaround and cost-effectiveness, with a predictive R^2 score of 0.954 using a graphical user interface (GUI) [264].

4.4 Numerical studies in FDM of parts

This subsection has summarized the overviews of different FDM 3D printing simulation analyses for various materials. The study has been conducted using commercially available software packages, including finite element analysis (FEA), Abaqus, LS-Dyna, ANSYS, and computational fluid dynamics (CFD). The mechanisms are relatively similar, as they use a common numerical simulation methodology, including solid model design, meshing (used to check the stability of the FEA), contact definitions, applying boundary conditions (to obtain the value and parameters at which the model tends to load applied) [265, 266]. The more detailed information about the analysis settings is shown as in Fig. 15.

According to study [267] printed PLA scaffolds have a higher ultimate compressive strength for lay-down patterns 0/90° and 0/60° with values of 133.9 MPa and 71.5 MPa, respectively, compared to ABS scaffolds which has an ultimate compressive strength of 67.667 MPa at 0/60° and 91.41 MPa at 0/90°. This is because PLA is more crystalline, higher molecular weight, and more regular and defined pore structure which is affected by directionality than other parameters that monitor its deformation.

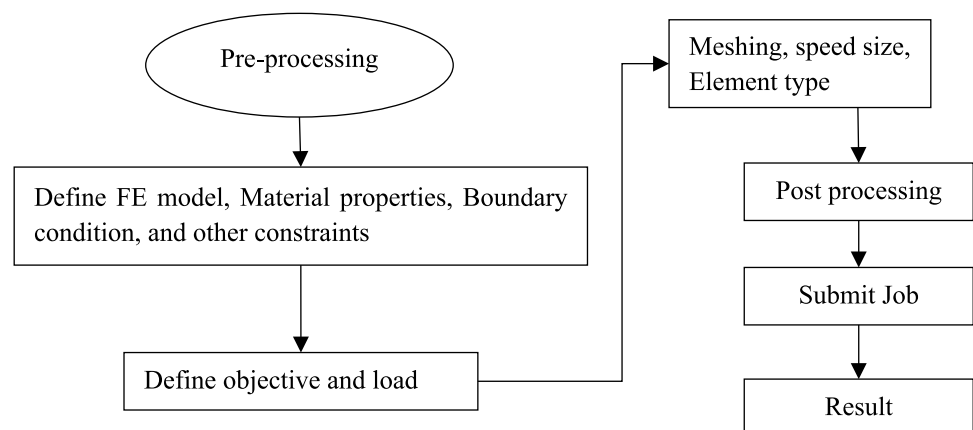
The same study, using finite element analysis [268], discovered that ABS-M30i scaffolds absorb more energy in body-centered cubic (BCC) lattice structure than in Schwarz primitive structure (TPMS) and simple cubic-double ring (SC-DRL) lattice structure. In terms of deformation, SC-DRL is higher than BCC and TPMS. Because the strain and plastic dissipation are not uniformly distributed across each layer and directionality of the filaments, the deformations (in mm) under FEM are 5.26 for SC-DLR, 4.56 for BCC, and 2.16 for SC-TPMS.

Another study [269] found a higher load-bearing capability and energy absorption capacity with double ring cubical lattice (SC-DRL) compared to both the honeycomb cubical structure (HCS) and the pyramidal cubical lattice structures (PCLS) when analyzed using Abaqus with ABS-M30i. Additionally, FEA indicated greater compressive deformation for all lattice types compared to the experimental results.

A study found a Von-Mises stress of 5.4 MPa for heel strikes and 10.8 MPa for push-off gait conditions for 3D-printed PLA socket. In addition, a maximum deformation of 0.74 mm for PLA socket and 2.66 mm for check socket, while a global deformation of 0.74 mm for PLA, 0.49 mm for composite, and 0.67 mm for polystyrene sockets was found. The observed tensile stress in the longitudinal direction with a forefoot loading of 19.4 MPa is closest to the tensile stress of 24.9 MPa for the 3D-printed PLA in the transverse direction. The study concludes that the lower-limb prosthesis can be manufactured using cheap, bio-based, and biodegradable PLA material, which would provide a cost-effective and eco-friendly alternative [270].

PLA filaments reinforced with 1% of hybrid montmorillonite (MMT) and multi-walled carbon nanotubes (CNTs) showed a 47.4% enhancement in tensile strength compared to virgin PLA utilizing ANSYS DMA, while the addition of 2% reduced the tensile strength due to nanofiller agglomeration. Tensile stresses of 20.42 MPa for PLA, 21.19 MPa for PLA with 1% CNT, 49.86 MPa for PLA-1% CNT-MMT,

Fig. 15 Different modules for numerical modelling and simulation



and 39.88 MPa for PLA-2% CNT-MMT were found. The MWCNT defined during the ANSYS has a purity of 99%, a length of 10 μm , a diameter of 5 to 15 nm, and a thickness of MMT is around 12 angstroms [271].

A cuboid part of PLA showed a maximum shrinkage deformation at the top, namely, shrinkage in the X direction and the initial layer on the bottom surface is attached to the XOY substrate. 0.0282 mm thermal shrinkage deformation (TSD) is found at a printing temperature (PT) of 190 °C and printing speed (PS) of 31 mm/s, and a maximum TSD of 0.0496 mm is found at PT of 219 °C and PS of 94 mm/s. In addition, support vector regression and cuckoo search for deformation control-oriented optimization enhances shrinkage distortion and shape exactness using FEM [272].

Using Simufact Forming software in FEM, study [273] revealed that the highest strength of 51.2 MPa is founded for clear PLA in YZ plane, which is higher than XY plane by 25%. For white PLA, a tensile strength of 36.4 MPa in the XY plane and 44.2 MPa in the YZ plane was found, while the composite PLA-wood filament had a tensile strength of 16.4 MPa, and the tensile strength for clear PLA varied by 3.12%. The XY plan exhibited a 43% reduction in tensile strength and a 49% reduction in elongation compared to samples printed from the same material in the YZ plane. Additionally, a composite PLA-wood demonstrated lower tensile strength than both white PLA samples and clear PLA. The cross-sectional shape and the specimen length were found to affect the tension, compression and bending of the printed PLA [274].

A study [275] found that both PLA and PLA with 40% wood fibers exhibited improved tensile resistance, although they also showed a decrease in strength and an increase in breakage. Using FEA, it was determined that the PLA composite had a fracture load of 4591.7 N for PLA and 1523.8 N for wood-reinforced PLA at 0°. The fracture load for PLA at 0° was recorded as 4591.7 N, whereas it was 3512.5 N for PLA at 90°. The addition of a small quantity of wood particles can increase the tensile resistance of the printed parts; however, incorporating a large number of wood particles results in reduced strength.

Incorporating a 90° or onyx buffer layer enhances the shear resistance of unidirectional CFRP, whereas 45° or 90° laminate fails to deliver the required strength in the single-lap joint configuration. In the instance of onyx with a 0° layer, onyx with a 90° layer, and 45° layers with a 45° interface, inadequate bonding was observed. Laminated carbon fiber exhibited a strength of 811 MPa, while laminated onyx showed a strength of 34.4 MPa. When comparing the strength of the 90° orientation of onyx combined with a 45° layer is noteworthy; however, the implementation of 45° layers is susceptible to sheer peel failure [276].

Utilizing numerical analysis, the same study increases the tensile strength of PLA-ABS functional graded (FG)

material made with FDM, while verifying the elastic modulus across different parameters. The ductility of the material improves with the increase in the thickness of the specimen. The FGM offers several advantages, including enhanced resistance to deformation, light weight, excellent recoverability, and improved toughness as demonstrated using Abacus [277].

The numerical simulation revealed that the ductility of PLA material increases with specimen thickness, showing kinetic energy values of 0.0186 J for 6 mm, 0.0544 J for 8 mm, and 0.0858 J for 10 mm according to the study [278] using the ANSYS explicit module, called LS-DYNA. Both the numerical simulation and experimental results exhibited a strong correlation.

A study [279] compares Lagrangian and arbitrary Lagrangian–Eulerian (ALE) FEA for tensile loading ABS specimens. While a yielding ALE formulation was proposed, it did not yield improved results due to the limitation of ABAQUS (standard) regarding element type selection, while Lagrangian demonstrated reduced CPU time. The implicit Lagrangian formulation recorded a maximum principal stress of 39.19 MPa, whereas the explicit Lagrangian formulation reported 39.72 MPa. For the ALE formulations, the implicit method found a maximum principal stress of 38.41 MPa, and the explicit method found 38.74 MPa. Furthermore, it was observed that as the number of mesh elements increased, the maximum principal stress also increased.

Finite element computation and experimental analysis reveal a complex deformation mechanism in microstructural implementation and printability of nylon [280], influenced by shearing, filament sequencing, compression, and tension in the raster and the frame. Printability is limited to a specific temperature range of 250 to 255 °C, with variations influenced by thermal cycling, porosity content, and mechanical performance. The analysis indicates that engineering stress levels can reach as high as 250 MPa, accompanied by a significant elongation at break ranging from 151 to 169%.

A maximum temperature fluctuation of 114.85 °C was observed for the grid path, while fluctuations for the concentric and zigzag paths in the transverse direction (L1) were significantly lower at 12.915 °C. In the height direction (L2), all three printing paths exhibited a uniform temperature fluctuation of 12.021 °C, indicating a negligible effect of printing path on the temperature field of PLA material. In addition, the zigzag path demonstrated a minimum stress of 0.187 MPa with uniform stress distribution, compared to the concentric path, which exhibited a higher stress of 2.14 MPa. Notably, temperature accumulation was observed at the centre of the printed part, raising the temperature to 178.93 °C. the zigzag printing path resulted in the smallest and most uniform stress distribution, while the deformation and complex thermal stress were attributed to the uneven distribution in the PLA part [281].

Another study [282] reported a tensile strength of 61.8 MPa when using a nozzle temperature of 220 °C and a nozzle velocity of 50 mm/s, for 3D printing with PLA incorporated 2% wt. GNP nanocomposites. In contrast, a lower tensile strength of 52.1 MPa was observed at a nozzle temperature of 180 °C and nozzle velocity of 30 mm/s as determined through numerical fluid flow on computational fluid dynamics FDM printing deposition. Results indicated that as the nozzle temperature and nozzle velocity increased, the thickness, area, and compactness of the deposited filament decreased. Additionally, under low temperature conditions, dimensional deviation decreased by 52.2%, 62.7%, and 68.3% in the X, Y, and Z directions, respectively. At the lowest nozzle temperature and high nozzle velocity, a cylindrical cross-section was observed, whereas higher nozzle temperatures paired with low nozzle velocities resulted in a flat cuboid with rounded edges.

The same study [283] focused on optimizing the design of a small hydrocyclone using CFD simulations on FDM. The results indicated that a 10 mm hydrocyclone with parabolic walls significantly improved separation performance, cost efficiency, and processing time compared to conventional conical wall designs. Specifically, the solids recovery and concentration ratio were increased by 10 percentage points and 0.2, respectively, when using a 0.5 g/L yeast suspension and at an equivalent pressure drop. The hydrocyclone with parabolic walls demonstrated a more considerable increase in solid recovery, particularly when the conic wall designs, when the vortex finder to spigot diameter ratios, were bigger. The operating pressures were found to be 110 kPa for parabolic hydrocyclones and 230 kPa for conic wall hydrocyclones, leading to the selection of the small hydrocyclone design for optimal performance.

The study [284], which used CFD simulation on 3D-printed PLA, found a tensile strength of 45.06 MPa with an R^2 of 97.29% and a 2.266% error at optimal cooling fan speed of 79.3%, extrusion temperature of 214.4 °C, printing speed of 75.9 mm/s, shell number 5, and raster width of 0.48 mm. Additionally, as the fan speed increases from 0 to 6500 rpm, the outlet velocity increases from 0 to 100 mm/s, while component velocity dropped from 100 to 80 K. The tensile strength initially increased with higher values of printing parameters (cooling fan speed, extrusion temperature, print speed, raster width, and shell number), but began to decrease beyond the second level of these parameters.

Additionally, a study [285] investigated the temperature distribution within a heated chamber subject to various 3D printing orientations and airflow rates using CFD results. It was found that increasing the temperature within a heated chamber significantly affects the quality of ABS printed in FDM. The average heat flux observed for each temperature variation in the specimen increases with increasing temperature in the chamber. When the heater is off, the average heat

flux is measured at 12.69 W/m², while it increases to 30.53 W/m² at 70 °C, 38.77 W/m² at 80 °C, 47.08 W/m² at 90 °C, and 60.83 W/m² at 100 °C. Under heating conditions, the temperature in the chamber rises and becomes evenly distributed; when the chamber reaches the target temperature (90 °C), the heat flux distribution creates a uniform layer in all printing orientations on the x, y, and z-axes. However, uneven temperature distribution within the chamber leads to problems such as warpage, shrinkage, geometric accuracy, and reduced mechanical properties.

Moreover, dual-temperature systems improve filament formation and deposition accuracy of poly(ϵ -caprolactone) to 50 μ m, demonstrating promise for the development of advanced tissue engineering scaffolds with interconnected pores and high surface area [286]. As printing speed increases from 0 to 2 mm/s, the filament width reduces from 350 to 50. In this dual temperature setup, viscosity showed a parabolic decrease while the shear rate showed a parabolic increase in the X direction; conversely, both viscosity and shear rate show linear behavior in the z direction. Viscosity changes from 1810 Pa.s to 3020 Pa.s under dual temperature control, whereas viscosity changes from 1550 to 2510 Pa.s with single temperature control over a printing duration of 3 s.

5 Significant contributions of the work and challenges of the FDM process

5.1 Significant contributions of the work

Extensive literature review The main contributions of this study are its methodical and comprehensive examination of the literature on the materials, process parameters, response metrics, numerical simulations, optimization techniques, and FDM applications. To provide a thorough overview of the current level of knowledge in this field, the study looks at the results of other studies. Researchers, engineers, and professionals can use the review resource to learn more about the FDM process mechanics, common materials, FDM 3D applications, and optimization techniques.

FDM materials and material enhancement The major contribution of the advancement of 3D printing is the development of new materials. For instance, new matrix composite materials have been developed that are stronger, more flexible, and more durable than pure materials. Examples of such materials are plastic composites, ceramic composites, metal composites, and metal-plastic composites. With the introduction of new materials, 3D printing machines have become faster and capable of printing with higher

resolution. This opened up new possibilities for the production of functional components in the field of biomedical and aerospace applications that require precision and intricate details in parts. In addition, performance improvement is another significant benefit of 3D-printed blending material. Because of their improved mechanical, chemical, and thermal resistance, the newer materials can be used in a greater variety of applications. This has created new opportunities for 3D printing in sectors like healthcare, automotive, and aerospace.

Identification of critical parameters and optimization strategies This is another noteworthy contribution which is the documentation of main process parameters that have a substantial influence on mechanical properties surface roughness and morphology the study explores the parameters including layer height build orientation and printing temperature significantly affecting the parts strength and durability in addition the study emphasizes materials made to order to specific printing conditions the study deliberates various techniques to improve the quality of the FDM-printed parts it explores the different designs of experiments including factorial design of experiments fractional experiments the Taguchi method central design of experiments and box new mathematical modelling techniques are addressed furthermore different optimization strategies were utilized to meet multiple performance objectives the method is used to fine-tune printing parameters enhance response metrics and strike a balance between various qualities.

Real-world applications By demonstrating how FDM process parameters affect part quality objectives, it offers scientists and professionals the benefits that are already available this data can be used by manufacturers to create printed parts of high quality and mechanical durability for several applications, including biomedical and customized products the FDM method facilitates the production system of scaffolds tissue engineering and medical implants the efficiency of the learning process and health care increases by using this process; it is also utilized for the fabrication of mechanical parts customized components and other fields of application.

Customization, eco-friendly, and recyclable materials The capacity of FDM to create eco-friendly and sustainable materials is another important characteristic of bio-based materials and recyclable thermoplastic materials, which lower waste and carbon emissions Furthermore, FDM enables the creation of intricate and highly customizable geometries that were previously unattainable through conventional manufacturing techniques it can facilitate the design of new materials and make FDM 3D printing an eco-friendly production method.

5.2 Significant challenges in FDM technology

The main goal of the FDM additive manufacturing process is to produce material by overcoming process difficulties. The fundamental challenges during the FDM process are described in this section. The part of FDM that influences mechanical properties requires a complex control system because of the significant effects of material compatibility, reinforcement distribution, cooling, and heating stages on mechanical integrity and material anisotropy when compared to the traditional manufacturing process. Furthermore, it is yet unknown what the optimal processing parameters are and how the blending ratio affects the final work pieces' characteristics. Moreover, FDM printing is less common for materials including metals, composites, ceramics, and mixed materials.

Certain FDM 3D printing processes only provide a limited range of printer parameter values; further level values are not available. Furthermore, some 3D FDM machines have a nozzle diameter of their own that sets a restriction on the number of contours and raster width. Additionally, each printed material has unique settings and parameters that affect the mechanical, surface, and thermal properties of the printed material. There are still no ideal circumstances for every kind of component and material. In contrast to traditional manufacturing techniques like injection molding, FDM is not appropriate for large-scale product fabrication and mass production. Shape memory polymers are used as the printing ingredients in 4D printing, a new manufacturing phase that has just been developed. Moreover, additive manufacturing technology is being adopted by 5D, 6D, and 7D printing technologies.

6 Future research directions

As we look towards the further development of FDM technology, various potential directions for additional study and investigation can be considered. Some of them are:

Advanced and multi-material investigations Numerous researchers looked into mechanical, thermal, and surface properties. In this context, future studies can concentrate on additional quality attributes such as stress–strain behavior at high-strain-rate loading conditions, interlayer bonding between different materials, the vibrational effect of the product, structural characteristics, dynamic properties, process planning of FDM, and the chemical makeup of products. Future studies will also focus on twin extrusion systems or concurrent multi-material development.

Advanced parameter control method The process mechanics and relationships between process parameters and response metrics are sophisticated, whereas the physical

constraints on FDM 3D printers, process parameters, material properties, and measurement methods, which collectively affect the quality of the printed component, are more straightforward. Future studies will explore the impact of process parameters on the quality of 3D-printed components made from multi-material and smart materials. Statistical methods, advanced mathematical modelling, and multi-objective optimization techniques will be used to determine optimal parameters. Future research will focus on using soft computing modelling and optimization techniques like ANN, fuzzy logic, TLBO, TOPSIS, multi-objective Jaya optimization, machine learning, and artificial intelligence to predict optimal process parameters based on desired quality attributes and material characteristics. FDM parameter optimization remains a popular research area due to its numerous parameters affecting quality features.

Standardization and quality control Component quality helps the manufacturing industries build standards for FDM printing. Due to its flexibility, capacity to make lot sizes, ability to print complex geometries, and ease and cheap, the component quality control method will be the future research direction. To meet standardization, a product can be printed using digital reverse engineering and recorded printed data.

Future practical applications Thought-out techniques for processing metals, ceramics, and polymers for use in manufacturing industries such as aerospace, automotive, and medicine. Metal matrix polymers like silicon carbide, alumina, titanium carbide, and carbon nanotubes are used in energy storage, aerospace, automotive, and tooling, while advanced smart materials like silicone elastomers, shape memory alloys, hydrogels, polyglycolic acid, polyvinylidene fluoride, and synthetic collagen are useful in the medical field, making them popular research topics. In reaction to environmental stimuli such as temperature, pressure, and light, these materials exhibit unique properties. Therefore, future studies focus on analyzing these materials and assessing their durability and effectiveness on FDM in various sectors.

Minimization of environmental effects In this changing world, climate change, global warming, and ozone layer depletion are major influences in every sector. Recycling biodegradable materials is therefore being adopted by sustainable manufacturing processes for environmental safety and consumption of energy. Therefore, FDM technology for supportable behaviors such as recycling and material reuse may be the focus of future studies. Additionally, future studies examine how the FDM 3D printing procedures affect the environment while considering different recycled materials to enhance the qualities of eco-friendly resources.

7 Conclusion

Additive manufacturing has attracted much attention in recent years since they are advanced to rapid production, custom-built design, and precise control over the dimension and porosity in automotive applications, medical fields, and personal use. In this regard, this review paper acknowledged numerous critical points in FDM, including the technologies of material blending, material properties, process parameter effect, mathematical modelling method, parameter optimization, numerical study, and future research direction on the FDM process. Numerous methods were used to make experimental designs and control variables, whereas optimization methods successfully optimized the process parameters and validated the effectiveness of the experimental result. In addition, the finite element analysis can be used as a basis for an experimental study. The potential outcomes of this review paper are the following:

- Numerous studies use numerical studies such as FEA, Abacus, and CFD for comparing the mechanical characteristics to the experimental result. Other studies use numerical simulation to obtain accurate deformation mechanisms, temperature degradation, and thermal stress distribution of FDM components during the building and cooling process which is inconvenient for experimental work. Furthermore, other studies analyzed porosity, tribological mechanisms, physical dimensions, surface roughness, and crystallinity.
- The research found very important data, highlighting that PLA, with its high stiffness and low plasticity, is ideal for prototypes and low-stress parts, while PETG offers a balance between flexibility and strength, suitable for functional components that require durability. ABS is known for its high-impact strength and ductility. Many research works have attempted to improve the mechanical properties and thermal characteristics of these materials, while other performance metrics and novel materials will be further research areas.
- Printing speed, nozzle temperature, infill density, build orientation, extrusion temperature, layer thickness, infill percentage, printing speed, and raster angle are the main influencing parameters on the quality and mechanical properties of the FDM printed part, while some process parameters, such as shell width, support structures, contour width, and number of contours, are less analyzed. However, further research comprehensively considered both parameters on the FDM printed part.
- Limited research is done on experimental design techniques such as full factorial, Taguchi, and CCD, while mathematical modelling and optimization techniques including Taguchi-GRA, RSM-GRA, ANN-GA, fuzzy

logic, and machine learning are studied. Thus, advanced optimization techniques have been undertaken for future studies considering multiple quality and process parameters of FDM to improve product quality, durability of 3D-printed parts, and efficient printing.

- Lastly, the method continues to drive advances in additive manufacturing, and rapid progress in material science, enabling customized solutions for diverse materials and applications in the field of 3D printing. It is a foundation for future research in additive manufacturing and material engineering that found improved mechanical characteristics and quality of printed components through the addition of fiber material, blending of pure biodegradable material, altering the lattice design, and heat treatment process, while it is necessary to advance more sophisticated technologies that fulfill environmentally friendly methods, enhance mechanical and thermal properties, maximize production efficacy, reduce time and cost.

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Data availability This work does not contain the author's own data.

Declarations

Conflict of interest The authors declare no competing interests.

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